

Type 205W

Multi-purpose water cooled motors/generators



User Manual

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Please keep for further reference.

Revision				
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02/2025	01.10	7.2	New material number for screw set with precoat (phase cable connection)	JB

This user manual has been created by us to the best of one's knowledge. Should you nevertheless discover any mistakes or lacks of clarity, don't hesitate to inform us. Furthermore, we would appreciate references and proposals. Please contact:

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Version 01.10

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Aachen, 7 May 2025

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1 Preface

1.1 This Manual

This manual describes the type 205W motors / generators in general as well as the application from delivery until disposal. Please keep this manual for further reference.

Read this manual carefully before you handle the machine. All instructions must be followed to guarantee a safe operation without any malfunctions.

The safety instructions must be followed at any time when working with the machine for your own safety, the safety of other persons and to avoid any property or environmental damages.

Figures in this manual are exemplary and might not show all features of a certain machine. All product names, trademarks and registered trademarks mentioned in this manual are property of their respective owners.

1.2 Guarantee and Liability

ENGIROs "General Terms and Conditions" apply. The currently valid version can be consulted here: <https://www.engiro.com/general-terms-and-conditions>

ENGIRO has regulated its third-party liability by subscription to liability insurance for consequential damages. Please contact ENGIRO for the currently insured sums.

The responsibility for determining the interfaces, for the application of the article of sale into its entire system, for the functionality of its entire system, as well as for its requirements including the absence of defects of customer-supplied items shall remain with the customer.

1.3 Textual Structures and Abbreviations

For the sake of readability in this manual only the term machine is used for any type 205W motor/generator, regardless of its use in the final application.

1.3.1 Textual structures

Function/item	Textual structure
Instructions	start with an arrow →
Lists	start with a point •
Enumerations	start with a number: 1) The right order of the steps must be respected

1.3.2 Abbreviations

Abbreviation	Meaning
AC	Alternating Current
DC	Direct Current
PCD	Pitch Circle Diameter
CW	Clockwise
CCW	Counterclockwise

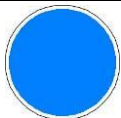
2 Safety Regulations

ENGIRO products may only be used for the applications specified in the associated technical documentation. Any used third-party products and components must be recommended or approved by ENGIRO. Proper and safe operation of the products requires proper transport, storage, assembly, installation, commissioning, operation, and maintenance. The permissible ambient conditions must be observed. The instructions in the accompanying documentation must be observed.

2.1 Safety Markings

This manual includes a consistent system of safety markings, which must be taken into consideration when operating any ENGIRO machine.

The meaning of safety markings will be signalled by form and colour.

Form	Colour	Meaning
	Safety colour red Contrasting colour white	Prohibition
	Safety colour yellow Contrasting colour black	Warning
	Safety colour blue Contrasting colour white	Mandatory measure

2.1.1 Harm Levels and Signal Words

The following signal words and safety symbols are to indicate dangers and dangerous situations of different harm levels. The indication of a higher harm level may include lower harm levels without the use of the corresponding marking. E.g., in case of DANGER, a WARNING might also apply.



DANGER

Death or grievous bodily harm **will occur** if the corresponding precautions will not be observed.



WARNING

Death or grievous bodily harm **may occur** if the corresponding precautions will not be observed.



CAUTION

Minor bodily harm may occur if the corresponding precautions will not be observed.



NOTICE

Material damages may occur if the corresponding precautions will not be observed.



Information

Here you will get information and reference notes so as you will be able to perform the following operations effective and in safe manner.



Additional Documentation

This symbol points out to further documents and instructions.

2.1.2 Additional Safety Symbols

For an easy perception and further classification of risks, the following safety symbols are used within this manual.

Symbol	Meaning	Symbol	Meaning
	Hazard area warning		Warning Risk of stumbling
	Warning Danger of crushing		Warning Slip hazard
	Warning Suspended load		Warning Dropping objects
	Hazardous voltages		Warning Hot surface
	Warning Flammable materials		Warning Objects harmful to health
	Cutting hazard to hands		Disconnect from the mains before starting to work.
	Wear safety gloves		Wear safety shoes
	Instructions for the disposal of waste		Don't touch! Housing is energized.

2.2 Five Safety Rules

For your personal safety and to avoid any property damage any operator must follow all safety relevant instructions as well as the 5 safety rules following EN 50110-1 for work in voltage-free states. Apply the following 5 rules before you start any work in the mentioned order:

- 1) Disconnect completely:
All electrical installation, including auxiliary circuits, must be disconnected from live parts on all poles.
- 2) Secure against re-connection.
- 3) Verify that the installation is dead.
- 4) Carry out grounding and short-circuiting.
- 5) Provide protection against adjacent live parts.

For recommissioning the machine after the work is finished, carry out the 5 steps in reverse order.

2.3 Safety Instructions

The basic premise for the efficient and failure-free installation and safe use of machines is the knowledge of basic safety regulations. This user manual gives instructions for the safe installation and operation of electric machines. Any additional internal health and safety regulations must be observed. Product information of third-party products and substances must be observed.



! DANGER

Electric hazard - danger of death from touching live parts.

- ➔ Any work on the electrical fittings should be carried out by a professional electrician.
- ➔ A direct connection between a machine and the high-voltage battery must not be established.
- ➔ Never work on live parts.
- ➔ Remove any loose, damaged, and scorched connections and scorched cables immediately.
- ➔ Cables must not be trapped or crushed. Be sure to lay all cables so that they cannot be damaged.
- ➔ Only work on the electric equipment when the attached load is switched off.

 WARNING



The machine safety will only be ensured if all safety devices are installed properly and are operating.

- Never operate a machine without safety devices.
- Only remove safety devices when the system is switched off.
- Install all safety devices after finishing repair work and check if they are working properly.

 WARNING



Fire hazard from overheating of the machine and/or power electronic controller.

The machine is actively cooled by a water-cooling system.

- Regularly check the cooling water circuit and the coolant level.
- Never operate the machine without functional temperature control.
- Chemical substances required for installation and commissioning might be highly flammable. Make sure you remove any flammable substance before you operate the machine.

 WARNING



Risks from unrecognized damages.

- Check the machines at regular intervals for damages.
- Only personnel authorized by the manufacturer may remedy damage.

 CAUTION



Slippery floor surface caused by spilled lubricants or leaked liquids.

- Clean the floor immediately when polluted.
- Dispose of the cleaning cloths into the provided collecting boxes.

 WARNING



Danger of burns due to hot surfaces!

- Never touch the running machine, the controller and/or other components of the machine during operation.
- Before starting any work, all components must have cooled down.

CAUTION



Never make any alterations, extensions, or conversions to the machine without the manufacturer's written approval. If external parts are bought it cannot be guaranteed that they have been designed and built-in compliance with the stress and safety requirements.

- Any conversion work to the machine requires the written confirmation of ENGIRO GmbH.
- Any machine parts that are not in a perfect condition must be replaced immediately.
- Use only original spare and wearing parts.



NOTICE



The disposal of lubricants and detergents is accurately ruled by environment protection laws and regulations.

- Deliver waste and used up lubricants to your hazardous waste point of acceptance.
- Slopped lubricants must be fixed using binding agents and disposed of as hazardous waste.
- Make provisions to collect slopped lubricants (sealed floor, collecting drip pans, collecting canvas covers).

2.4 Staff Training and Duty of Care

Safety at work depends strongly on the attention, prevention, and reasoning of all involved persons. Beside the general safety instructions, please take care of

- general work safety regulations in the country of deployment.
- specific regulations of the operator and final application.
- specific agreements with the operator.
- safety instructions delivered with the machine.
- safety instructions delivered with any third-party accessory meant for the use with the machine.
- safety symbols on the machine as well as any accessory and its packaging.

For example, in Germany the responsible electrician according to DIN VDE 1000-10:2009-01 section 3.1 and 5.3 must be involved in any work on electric parts.

2.4.1 General

A machine may never be operated by personnel

- who is under the influence of substances which have an impact on its reaction,
- who is not able for health reasons to operate the machine.

Personnel under education, instruction or training may only work with the machine under the permanent supervision of an experienced and qualified person.



Information

The mounting instructions must be available permanently at the machine's place of installation. The place where the mounting instructions are filed must be known to involved employees.

2.4.2 Duty of Care of Staff

Any staff working with the machine must.

- have read and understood the mounting instructions.
- have been briefed in the mode of operation of the machine.
- know how to perform the particular operations on the machine.
- be in good health to install the machine.

2.4.3 Staff Qualification

All work on machines may only be performed by qualified persons. Qualified persons in the sense of this manual are persons who are able.

- to judge the works which have been delegated to them due to their professional training, skills, and experience as well as the knowledge of the corresponding regulations,
- to identify possible hazards and
- to take the necessary procedures to remedy risks of accident.

Job	Qualification
Assembly/Set up	Persons with technical training
Checks	Qualified persons
Work on electric parts	Qualified Electrician in accordance with company internal regulations as well as national regulations and law.

3 System Description

3.1 Intended Use

Type 205W machines are multi-purpose rotating electrical machines. These machines are suitable for a wide range of drive applications in combination with a power electronic inverter. They provide high output powers at low volume and weight, high reliability, and lifetime.

The machines are intended to be used as energy converter in combination with suitable power electronic controllers. The integration into a greater sphere of application is the responsibility of the user.

Any other use of the machines not in line with its original purpose is prohibited.

The correct use of the machine also includes:

- The observance of all warnings in the user manual.
- The consideration of the maximum mechanical and electrical ratings of the specific machine
- The consideration of all legal documents related to the final product.

3.2 General Description Type 205W Motor/Generator

To cover a wide range of torque and power ratings, the design of the 205W is modular. Power and torque are varied by varying the active length and thus the housing length, see the Figure 1 & Figure 2. For the specific housing (and overall) length, please refer to the specific machine datasheet, available online.

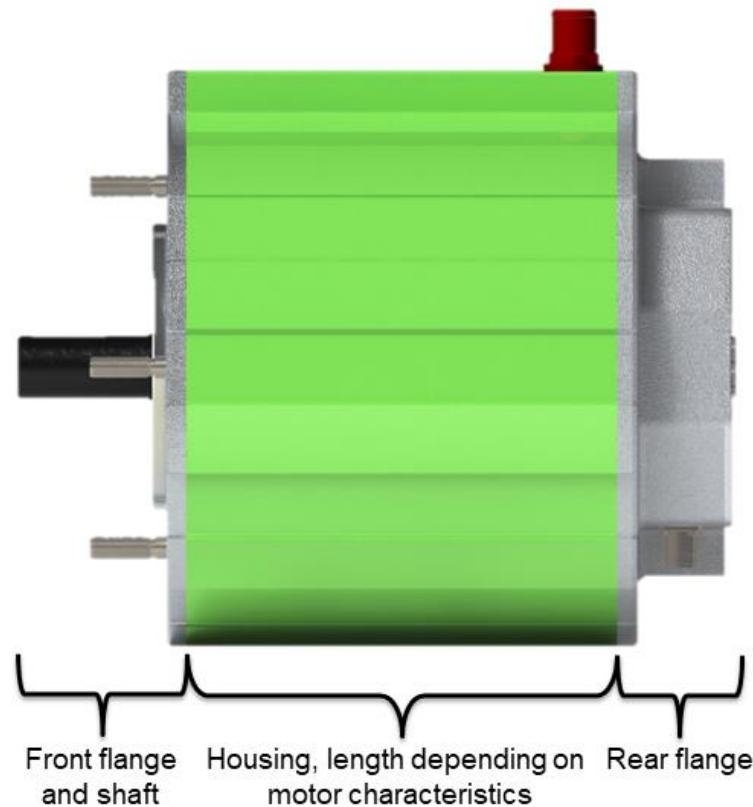


Figure 1: 205W composition

For integration of the 205W into a range of systems, the front flange and shaft can be selected by the customer. For available flanges and shafts, refer to sections 5.4 and 5.5. To be able to distinguish the machine types, each version has a unique part number for a corresponding motor type:

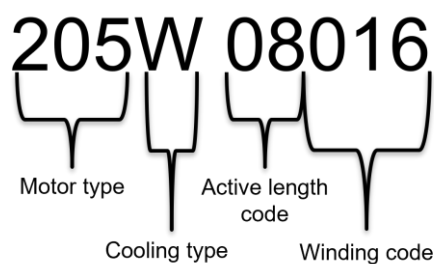


Figure 2: Motor type

For the position of the part number and serial number of the motor see chapter 9.

3.3 Maximum Operation Ratings

The mechanical and electrical system of the type 205W series is designed for the following operation range:

Parameter	Maximal Value	Unit
Mechanical speed	8000 ¹⁾	rpm
Maximum dc supply voltage for LV motors (in addition to equation 1)	200	Vdc
Maximum dc supply voltage for HV motors (in addition to equation 1)	850	Vdc
Acceleration	project dependent	rpm/s
Inverter switching frequency	8 - 12 ²⁾	kHz
Ambient temperature	-20 ... +85 ³⁾	°C

- 1) project/motor configuration depending; see datasheet
- 2) Other frequencies can have significant impact on motor performance. For operating conditions outside the mentioned limits please contact ENGIRO support.
- 3) Linear derating of phase current for ambient temperatures above 60°C to 0 Arms at 85°C

Voltage Class	Max. Surge Voltage in V Phase to Phase
LV motors	1200
HV motors	3000



Information

The motor was tested for operation at sea level. It is designed for operation up to 4000 m above sea level. Make sure the cooling system is adopted to lower air density.

Operating a machine at a different voltage than stated in the datasheet is possible. However, high nominal mechanical speeds might result from a too high dc-supply voltage. Therefore, it is advised to limit the maximal dc-supply voltage to:

$$V_{dc,max} = 8 \cdot U / n_{peak} \quad (1)$$

The corresponding peak voltage per speed constant U/n_{peak} -value can be found in the datasheet of the specific 205W variant. In addition, the maximum DC-supply voltage in the table above only applies if the inverter is based on Si-IGBT technology and is strongly dependent on the AC cable length (see Figure 3 and Figure 4). In general, AC cables length should be as short as possible.

For operating conditions outside the mentioned limits, please contact ENGIRO support.



Consult the corresponding datasheet for the specific limits of a certain type 205W variant. The latest version of datasheets can be found here: <https://www.engiro.com/products/motors-and-generators>

Max. Cable Length for LV motors

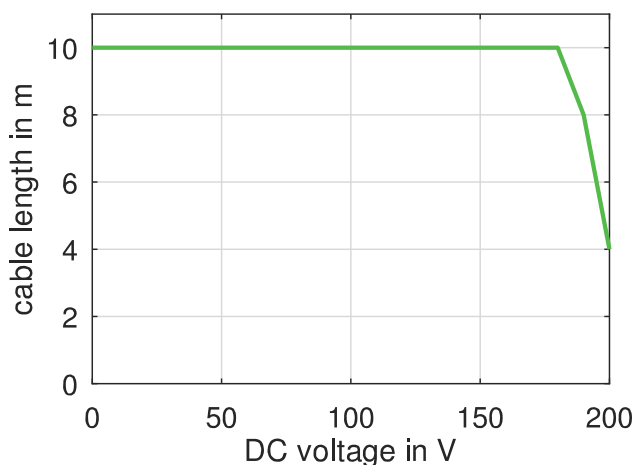


Figure 3: Max. cable length for LV motors at different voltages.

Max. Cable Length for HV motors

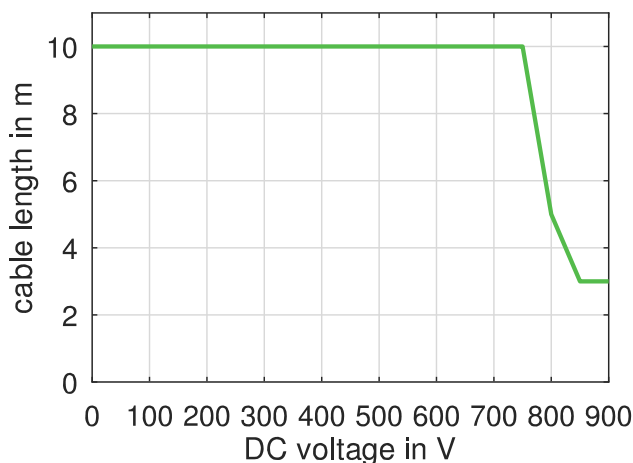


Figure 4: Max. cable length for HV motors at different voltages.

3.4 Lifetime

The nominal lifetime of the machine is 10.000 h if all operational limits are respected (e.g., axial/radial loads, thermal limits, switching frequency). Please contact ENGIRO if your application requires longer lifetimes.

3.5 Environmental conditions

ENGIRO motors are designed and tested to withstand certain mechanical, chemical and thermal environmental conditions. Please contact ENGIRO for further information. The 205W machines are built out of aluminum alloys. Shafts are supplied with corrosion protection.

For information on the machine's IP ratings and further certifications please see the relevant motor datasheet.

Please note that an IP rating is only valid if the machine is installed with suitable cable glands and an appropriate sealed interface at the drive side of the motor (flange and/or shaft). Please contact ENGIRO for further questions.

4 Transportation and Installation Preparation

4.1 Transportation and Storage

The machines should always be transported and stored in their original packaging. The machine must stand in a horizontal orientation and any external loads should be avoided.

Storage is limited to two years, as increased storage time can have a negative effect on the bearing grease. When storage time exceeds two years, turn motor shaft 3 ¼ turns to make sure grease is redistributed and bearing ring changes position.



Information

If the machine is stored prior to installation, the place of storage must be dry and protected from direct sunlight.

Room temperature should be 10°C – 25°C.

Transportation temperature should be -20°C – 85°C.

4.2 Unboxing

During shipping the machines are protected by a cardboard box lined with paper honeycomb sandwich panels. A PE bag is added to protect the machine against dust or moisture. After unboxing please

- ➔ Check the completeness of the delivery.
- ➔ If the delivery is not complete, please contact ENGIRO GmbH, immediately.

Please dispose all packing materials according to local regulations.

4.3 Handling Remarks

Machines with production date 02.2019 (Date code >1902DD) and later are suitable to be lifted using a lifting eyebolt DIN 580, size M8 (see figure 5). The production date is printed on the engine (see chapter 9). If this information is not available due to label data changes, please contact ENGIRO. The eyebolt is not delivered with the machine. Please take care that an eyebolt of the correct specifications is used.

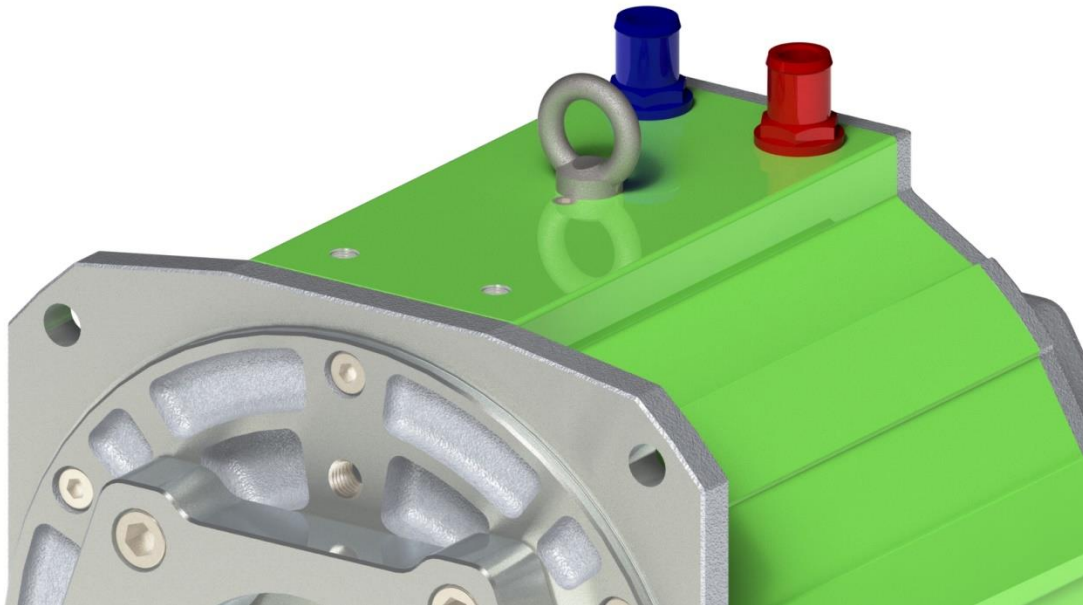


Figure 5: Lifting eyebolt installed onto 205W

Machines produced up to the above-mentioned production date should be lifted using a hoisting strap, placed around the engine. Alternatively, lifting aids can be mounted to the machine mounting points. Never lift the machine at the shaft or the cables.



NOTICE

Dropping will result in damage to the machine. Provide safe transport of the machine to its place of installation.



CAUTION

Risk of crushing by dropping the machine. Always consider the weight of the specific machine. Never lift a machine with less than 2 persons.



Consult the corresponding datasheet for the specific data of a certain type 205W machine. The latest version of datasheets can be found here: <https://www.engiro.com/products/motors-and-generators>

4.3.1 Housing bores

Machines with production date 11.2020 (Date code >2011DD) and later have two M10 threaded holes on the housing due to manufacturing reasons (see figure 6). The production date is printed on the engine (see chapter 9). If this information is not available due to label data changes, please contact ENGIRO. These holes are expressly NOT intended for handling or mounting the machine. Furthermore, no additional attachments should be fitted over these threaded holes.

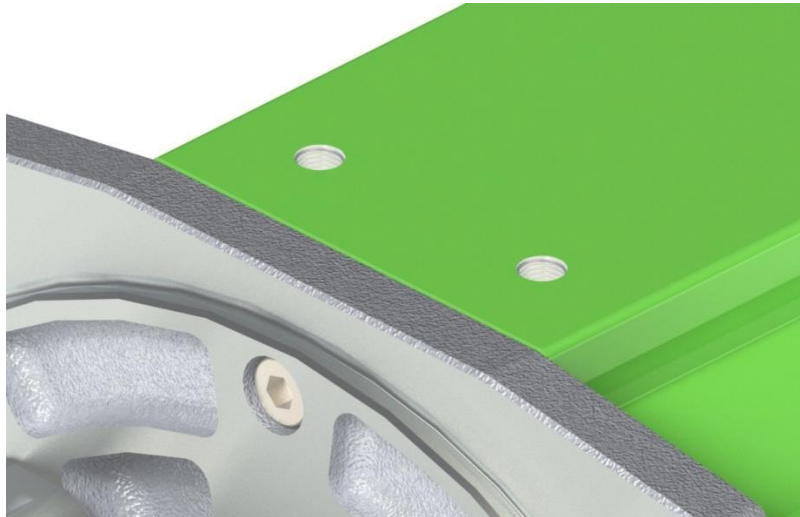


Figure 6: Housing bores due to manufacturing reasons

4.4 Initial Visual Inspection

Before installation, several checks should be carried out to make sure the machine is ready for operation:

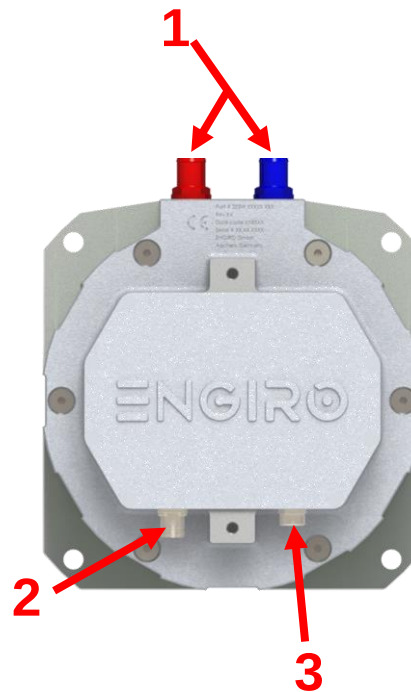
- ➔ Check for visual damage on any of the machine components.
- ➔ If possible, turn the machine shaft by hand. This might require some effort due to the permanent magnets inside the machine. Make sure that the shaft runs freely without abnormal noises like scraping or clicking.
- ➔ Check the inside of the connection box for moisture and make sure the cover seal is correctly installed and undamaged.

If any of these checks reveal a problem or a component is missing, please contact ENGIRO and do not install or operate the machine.

5 Mechanical Installation

5.1 General Motor-Overview

All motors are equipped with two 19mm ($\frac{3}{4}$ ") hose barbs (1) to connect the motor to a cooling system. The sensor connector (2) is located on the lower left of the rear flange facing downward. On the right side a pressure venting plug (3) is seated. 205W motors with a six-phase winding configuration are equipped with a second sensor connector in place of the air vent. In this case the air vent is relocated to the top right of the rear flange facing upward.



- 1 Water cooling hose barbs
- 2 Sensor connector
- 3 Pressure venting plug

5.2 Machine Installation Position

The machine must be positioned in a way that air will easily bleed out of the cooling circuit. For this reason, a certain range of machine positions shown in Figure 7 and Figure 8 are preferred.

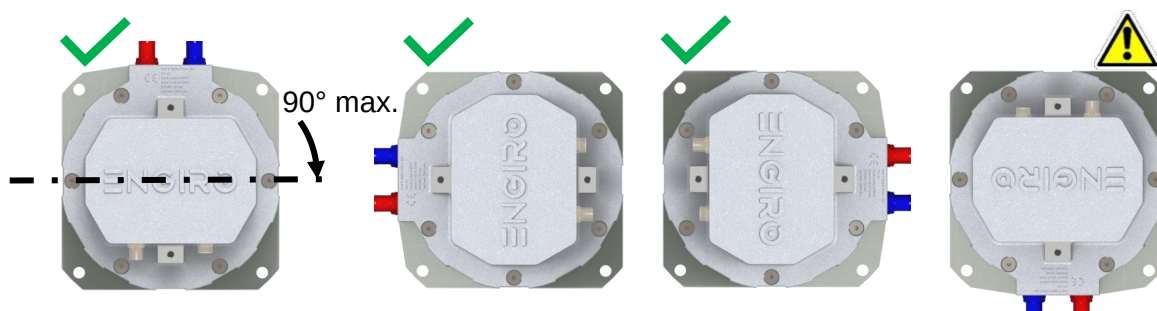


Figure 7: Rotational mounting position

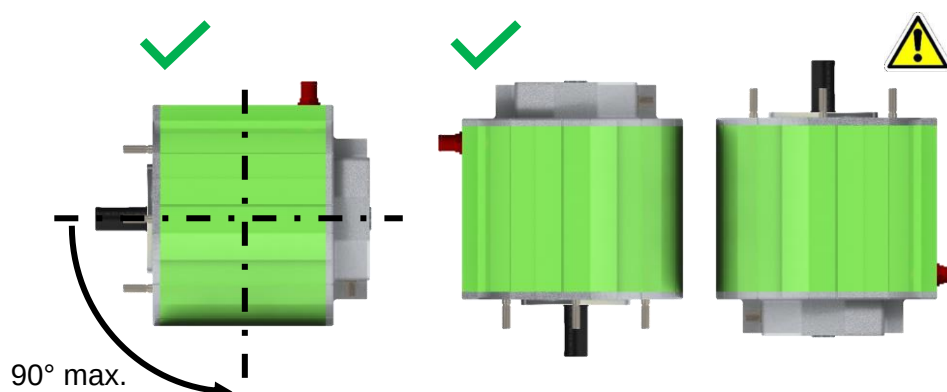


Figure 8: Shaft orientation



NOTICE

If the machine must be installed in an unfavourable position, special precautions should be taken to make sure there is no air in the machine cooling circuit: e.g. filling the cooling system in the horizontal position and then turning the machine to its final position makes sure all the air is bled out of the system.

The design, type of installation and the position of the terminal box of each motor combination is classified according to DIN EN 60034-7. The corresponding designations are given in the table below:

Motor type	IM-Code
205W_XXXXX_C1D1X	IM3049R
205W_XXXXX_C1G1X	IM3049R
205W_XXXXX_S1H1X	IM3649R
205W_XXXXX_S1S1X	IM3641R
205W_XXXXX_B1C1X	IM3041R
205W_XXXXX_S1E1X	IM3649R
205W_XXXXX_D1J1X	IM3049R
205W_XXXXX_D1L1X	IM3041R

IM-Codes for non-listed flange/shaft combinations can be requested from Engiro.

5.3 Bleeding or emptying the cooling system

The coolant can be drained from the cooling system by rotating the machine upside down.

5.4 Mounting Flanges

The Datasheet specifies the exact flange/shaft coding corresponding to your motor part number.

5.4.1 Flange Type B1

Machines equipped with a flange type B1 have a mounting flange with four holes $\varnothing 13.5$ mm, suitable for M12 screws. These are positioned on a PCD of 300 mm with 90° spacing (see Figure 9 & Figure 10).

Flange type B1 was designed for machines replacing hydraulic motors. The pilot diameter and bolt circle correspond to those on hydraulic pumps type ANSI 127-4 / SAE C. For mounting the machine to hydraulic pump mounts, use screws size M12x1.75 of adequate length and minimum strength class 8.8. A $\varnothing 127$ mm centering shoulder is provided for centering a hydraulic pump.

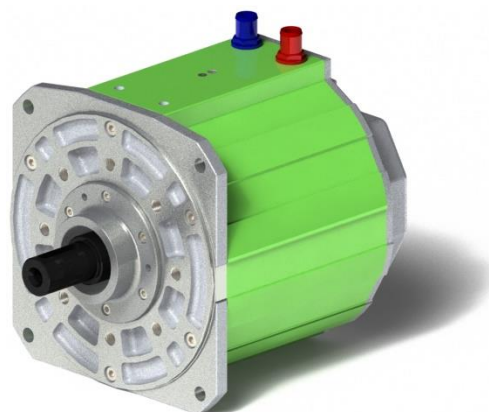


Figure 9: Flange type B1 with shaft type C1

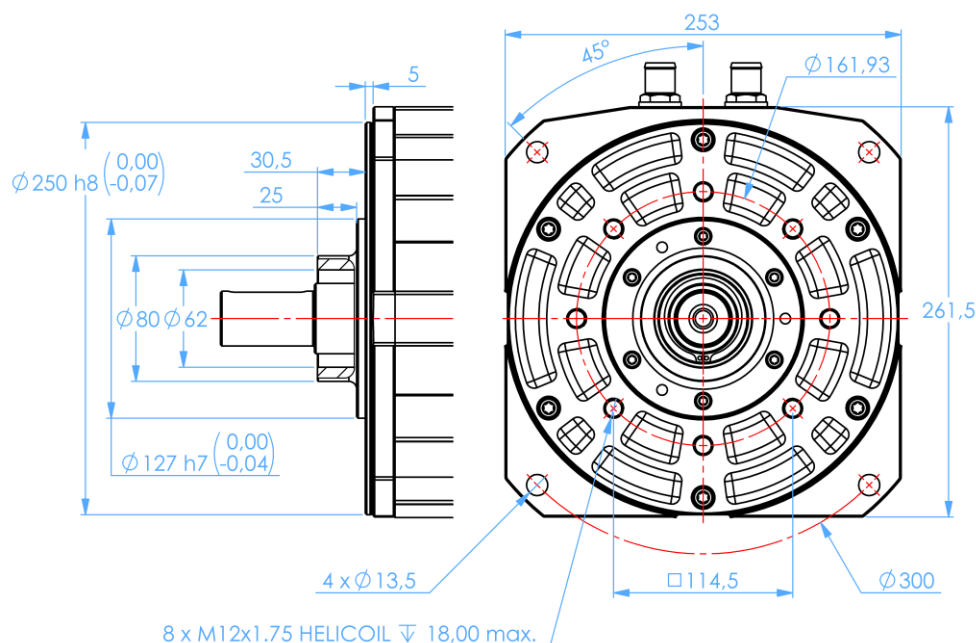


Figure 10: Dimensions flange type B1

Machine mounting for flange type B1

Type	Screws and nuts DIN 934 M12 min. strength class 8.8 / 8
Thread locking	Schnorr safety washers or Loctite glue
Tightening torque	68 Nm

5.4.2 Flange Type C1

Machines equipped with a flange type C1 have a mounting flange with four holes $\varnothing 13.5$ mm, suitable for M12 screws. These are positioned on a PCD of 300 mm with 90° spacing (see Figure 11 & Figure 12).

Flange type C1 (typically combined with shaft type D1) is designed to fit to the housing flange of a SAE type B / ANSI 101-2 pump. Two threaded holes M12 allow a hydraulic pump to be mounted. The complete set can then be mounted at the outer flange. Depending on the weight of the pump, it could be necessary to provide an additional support at the rear of the pump.

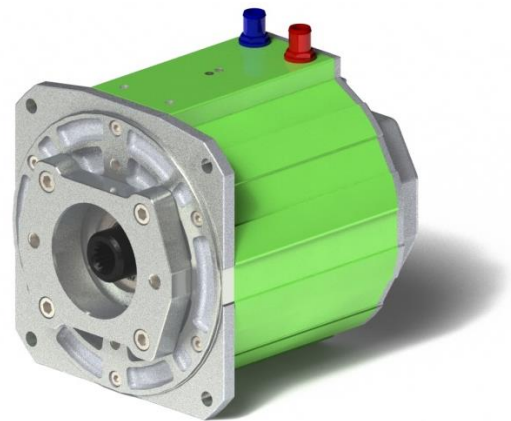


Figure 11: Flange type C1 with shaft type D1

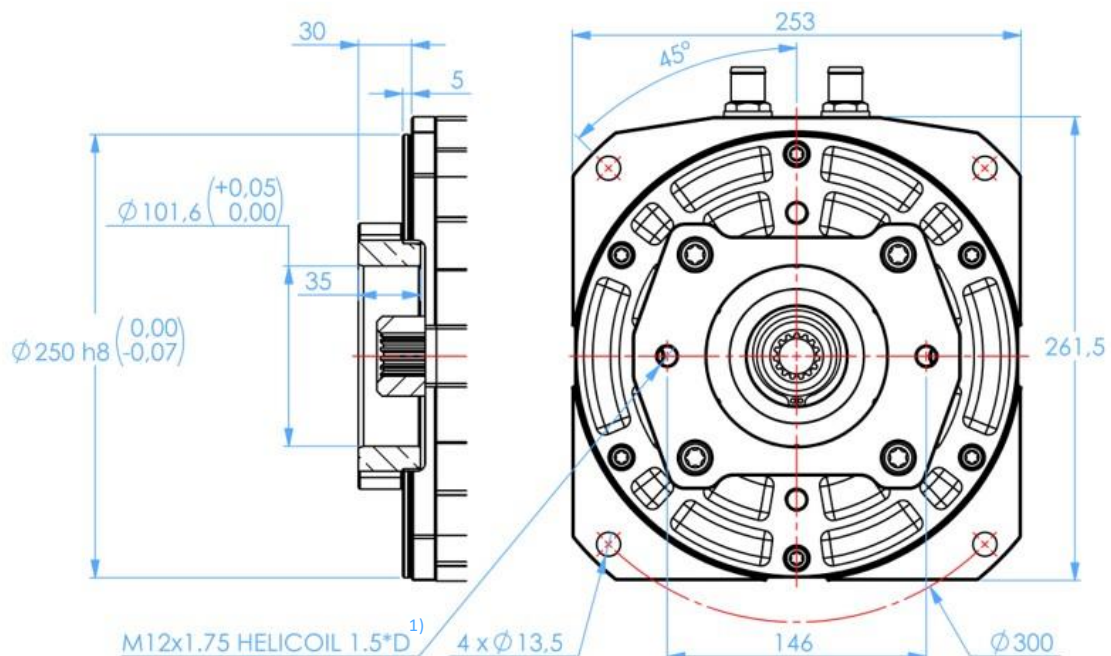


Figure 12: Flange type C1

- 1) Machines with C1-Flange and a revision number smaller than Rev15 have an M14 Helicoil 1,5*D. Revision number is printed on each machine on the rear flange below the water-cooling hose barbs as shown in Figure 56.

Machine mounting for flange type C1

Type	Screws DIN 912 M12 min. strength class 8.8
Thread locking	Schnorr safety washers or Loctite glue
Tightening torque	68 Nm

Flange Type D1

Machines equipped with a flange type D1 have a mounting flange with four holes $\varnothing 13.5$ mm, suitable for M12 screws. These are positioned on a PCD of 300 mm with 90° spacing (see Figure 13 & Figure 14).

Flange type D1 was designed for machines replacing hydraulic motors. The pilot diameter and bolt circle correspond to those on hydraulic pumps type ANSI 127-4 / SAE C. For mounting the machine to hydraulic pump mounts, use screws size M12x1.75 of adequate length and minimum strength class 8.8. A $\varnothing 127$ mm centering hole is provided for centering a hydraulic pump.

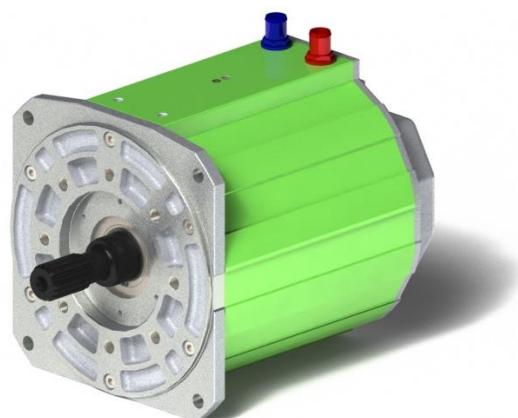


Figure 13: Flange type D1 with shaft type J1

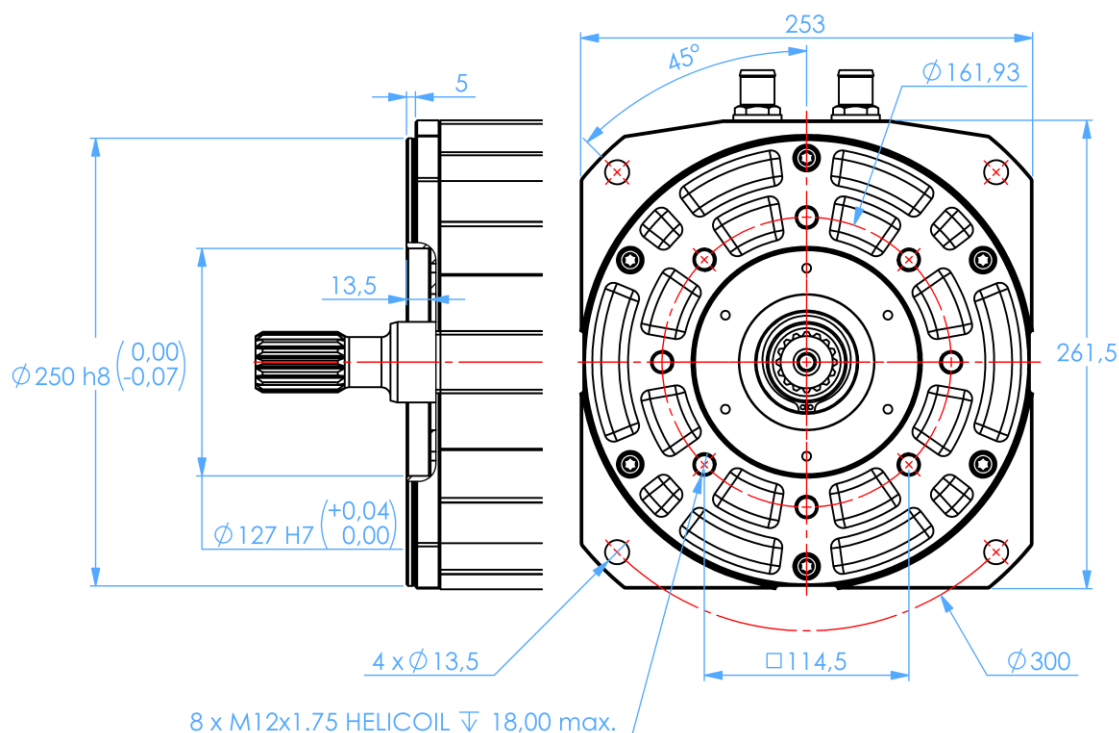


Figure 14: Dimensions flange type D1

Machine mounting for flange type D1

Type	Screws and nuts DIN 934 M12 min. strength class 8.8 / 8
Thread locking	Schnorr safety washers or Loctite glue
Tightening torque	68 Nm

5.4.3 Flange Type S1

Machines equipped with a flange type S1 have four studs M10, on a PCD of 165 mm. These studs are meant for mounting the machine and can support the full weight of the machine. For applications where exact alignment of the machine is important, a pilot diameter of 130 mm is machined onto the flange (see Figure 15 & Figure 16).

For mounting the machine to a flange nuts type DIN 934 or similar should be chosen, with a minimal strength class of 8. The required tightening torque for these nuts is 40 Nm. To prevent loosening in service, a proper method of thread locking should be applied; this can be either thread locking glue or safety washers. If the studs are damaged, please refer to section 12.2 for replacement information.

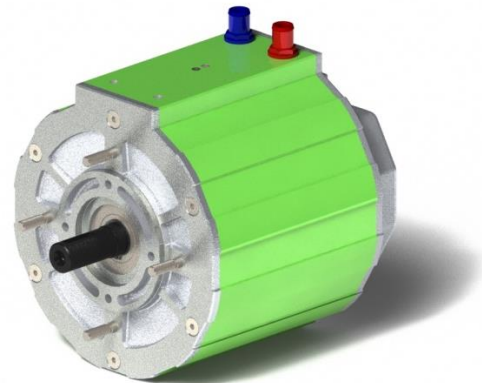


Figure 15: Flange type S1 with shaft type S1

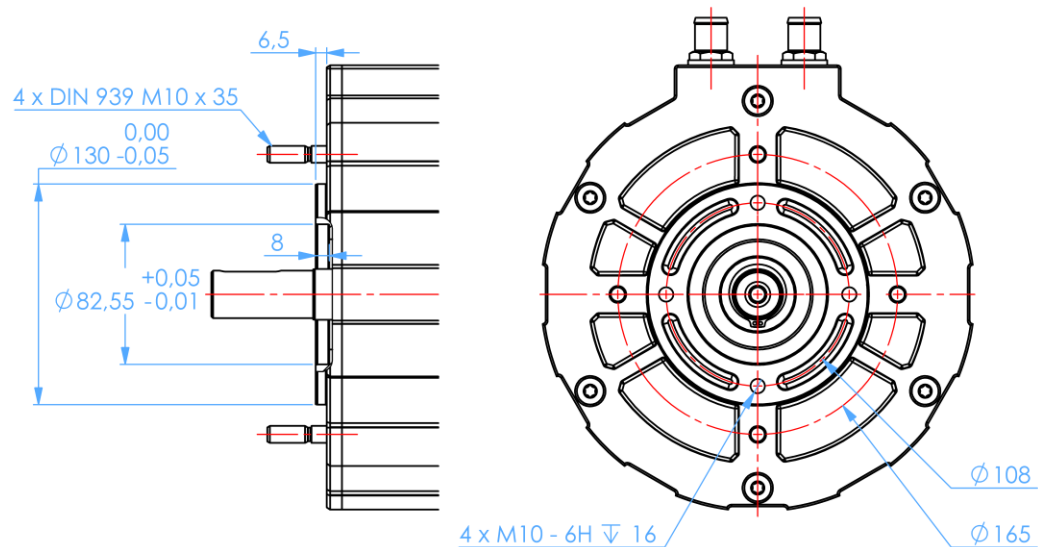


Figure 16: Dimensions flange type S1

Machine mounting for flange type S1	
Type	Nuts DIN 934 M10 min. strength class 8
Thread locking	Schnorr safety washers or Loctite glue
Tightening torque	40 Nm

In addition to the machine mounting studs, all type S1 flanges have a pilot diameter of 82,55 mm and an extra set of M10 threaded holes serving as a hydraulic pump mount. These holes are only meant for mounting a pump to the machine, not for supporting the machines weight. In hydraulic pump applications the motor-pump set must always be mounted to its carrier using the studs at PCD 165 mm.

5.4.4 Rear Flange

For some applications a support at the backside of the machine may be required. For this reason, two mounting points (M10 threaded holes) are located at the rear flange. These threaded holes can be used for either grounding or motor mounting (for grounding see section 7.1); this should be considered when designing the rear machine mount (see Figure 17 & Figure 18).

Provided the minimal screw-in depth of 16 mm is used, the machine mounting screws should be tightened to 40 Nm.

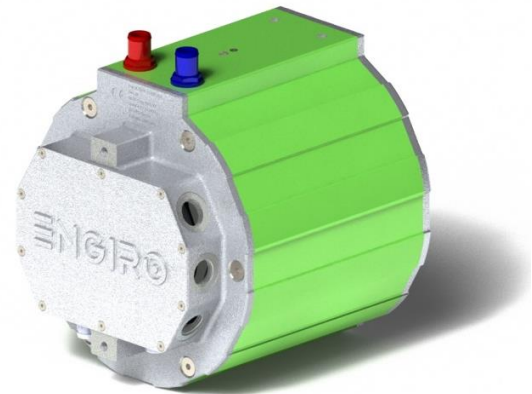


Figure 17: Rear flange

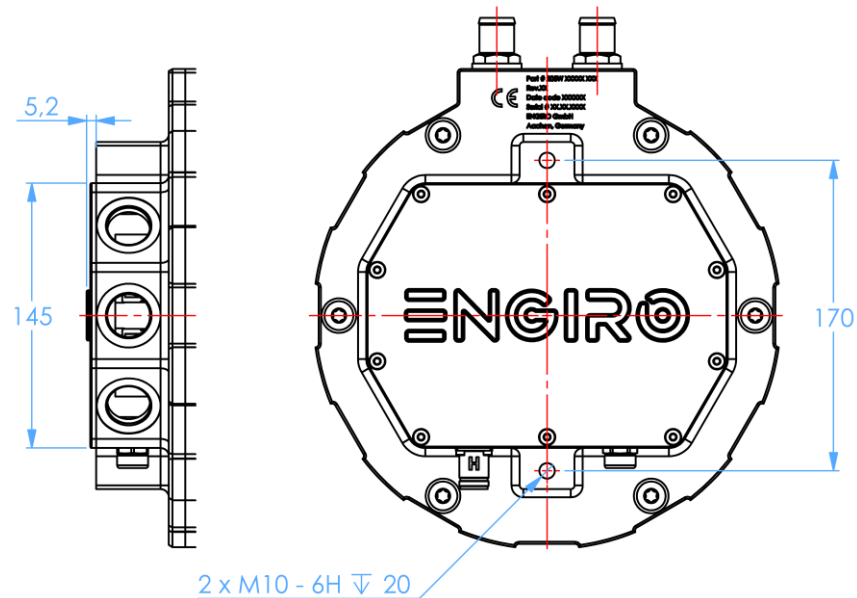


Figure 18: Dimensions Rear flange

Machine mounting rear flange	
Type	Screws DIN 912 M10 min. strength class 8.8
Thread locking	Schnorr safety washers or Loctite glue
Tightening torque	40 Nm

5.5 Shafts

The Datasheet specifies the exact flange/shaft coding corresponding to your motor part number.

5.5.1 Allowable Shaft Loads

In the operation of electrical machines, various combinations of forces and torques usually occur on the shaft. In order to achieve a service life of 10.000 hours, it is important to know the loads that occur and to limit them.

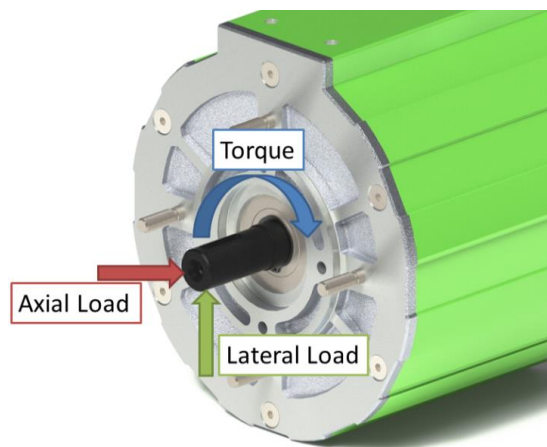


Figure 19: Torque and load definition

The individual shafts are designed to withstand the maximum torque of the corresponding motor variant, considering an application factor of $K_A=1.25$ (according to DIN 3990). If the expected torques are higher, approval by ENGIRO is required.

Lateral and axial forces affect the radial ball bearings. To avoid bearing damage, corresponding loads should therefore not be exceeded. The standards DIN ISO 76 and DIN ISO 281 serve as a basis for calculation. A distinction is made between the two cases for static and dynamic loading.

The static load case exists if the bearing under consideration does not perform any rotational motion. The verification of static load safety is intended to prevent individual elements in the bearing from being plastically deformed due to high loads occurring. Such high loads on the stationary bearing can occur, for example, when belt drives are installed. Unlike the axial loads, the permissible lateral loads depend on the selected shaft type and the machine length. The maximum permissible equivalent static bearing load is for all motor variants:

$$P_0 = 0,6 \cdot F_{L0} + 0,5 \cdot F_A \leq 10.000N \quad (2)$$

The load F_{L0} is a substitute load that is applied directly to the bearing. It results from the actual lateral shaft load, considering an application factor.

$$F_{L0} = K \cdot F_L \quad (3)$$

It is assumed that the lateral load acts on the outermost shaft end. This results in the following application factors K for the different shaft types and machine lengths:

		Active length code				
		04	08	12	16	20
Shaft type	C1	1,86	1,66	1,53	1,44	1,38
	D1					
	E1	Shafts with internal and external splines should not be exposed to lateral and axial forces! However, if these loads cannot be excluded during operation, a verification by ENGIRO is required.				
	G1					
	H1					
	J1					
	L1	1,55	1,42	1,35	1,28	1,24
	S1	1,55	1,42	1,35	1,28	-

The verification of static load safety excludes damage to the bearing at speeds $<10 \text{ min}^{-1}$. Operation of the machine at higher speeds and increased bearing loads requires a calculation of the dynamic load carrying capacity. Figure 20 and Figure 21 both show the allowable bearing loads for different operational speeds of the machine to maintain a service lifetime of 10.000 hours.

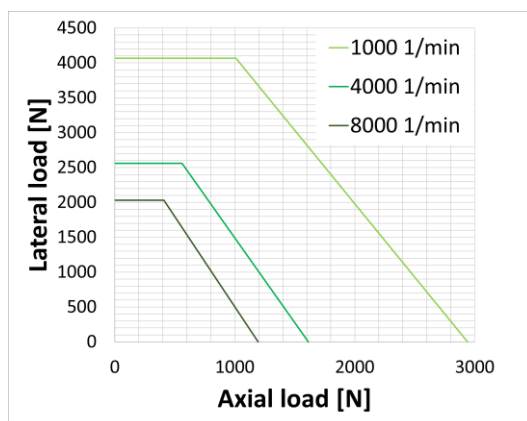


Figure 20: Front bearing
(Flange type B1, C1, D1)

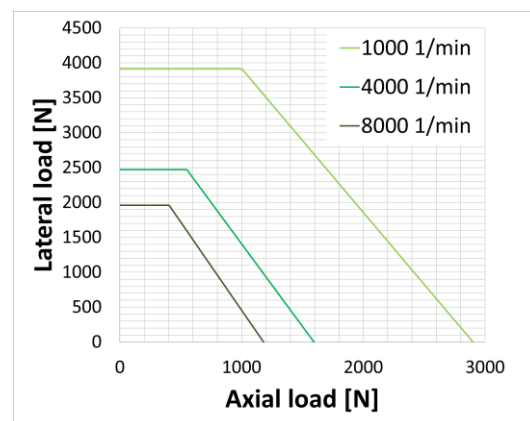


Figure 21: Front bearing
(Flange type S1)

If you are unsure whether the expected loads will significantly reduce the bearing life, please contact ENGIRO.



NOTICE

To maintain a bearing lifetime of 10.000 h please avoid higher loads.

5.5.2 Shaft Type C1 – 35 mm Cylindrical Shaft

The shaft type C1 is equipped with a cylindrical shaft of 35 mm diameter and 10 mm keyway (Key: DIN 6885 A 10 x 8 x 50). The combination with flange B1 substitutes hydraulic pumps and machines of type SAE C / ANSI 127-2. The specifications of all customer-relevant interfaces of the C1-shaft are shown Figure 23 as an example in combination with a B1-flange.

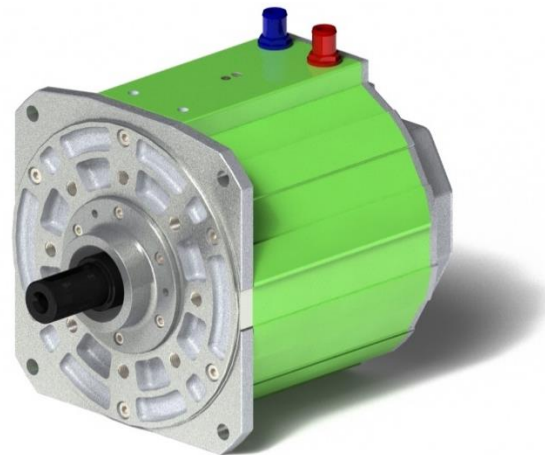


Figure 22: Shaft type C1 with flange type B1

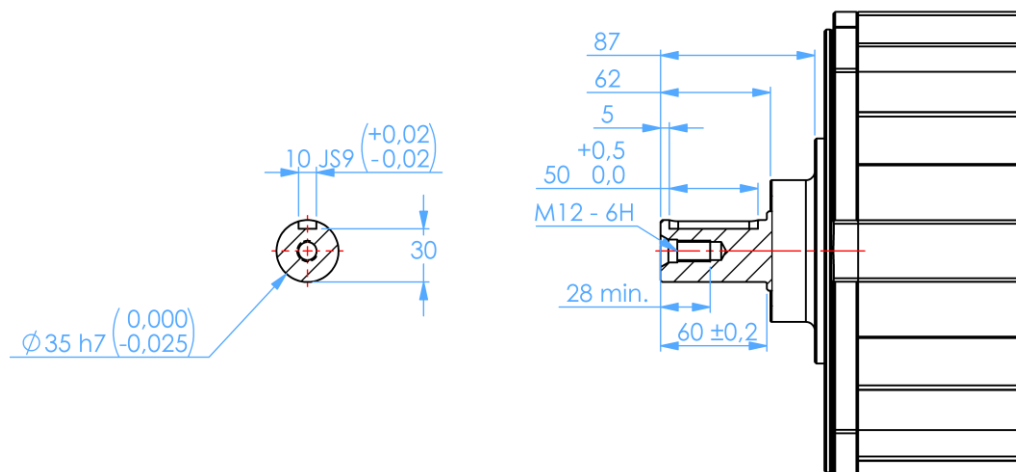


Figure 23: Dimensions shaft type C1



NOTICE

Please make sure that the mechanical load is distributed evenly along the length of the keyway. Avoid large torque impulses for shafts with keyways.

5.5.3 Shaft Type D1 – Internal Spline 15T

Shaft type D1 equipped with internal spline (ANSI B92.1 15T 16/32DP 30°) and is typically combined with housing flange C1. These machines are suited to be coupled with a hydraulic pump with Flange SAE type B / ANSI 101-2 and shaft SAE Type B-B / ANSI 25-4. The internal spline is hardened. The specifications of all customer-relevant interfaces of the D1-shaft are shown in Figure 25 as an example in combination with a C1-flange.

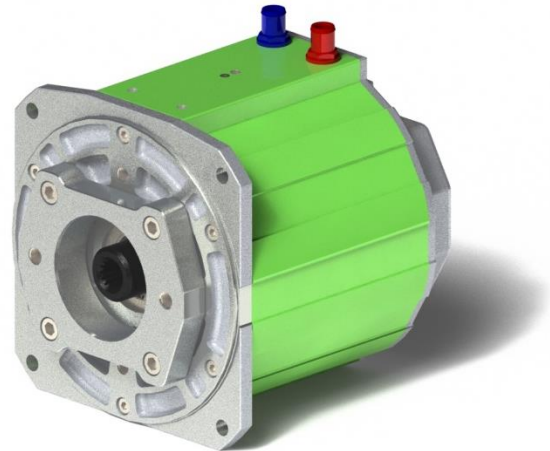


Figure 24: Shaft type D1 with flange type C1

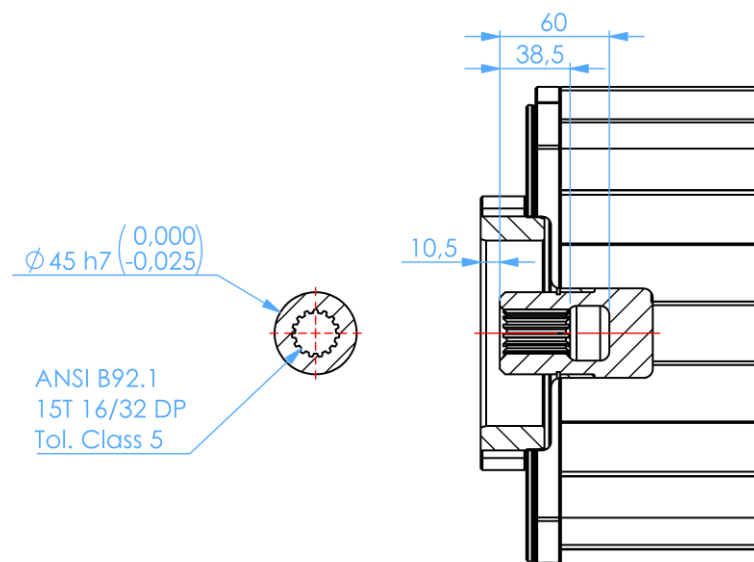


Figure 25: Dimensions shaft type D1

NOTICE



The motors are supplied with corrosion protection on the shaft end. This is not intended as a lubricant for the spline shaft. Spline couplings require specific lubrication according to the characteristics of the application and ENGIRO recommends the consultation of the lubrication suppliers to ensure proper selection and definition of re-lubrication intervals.

5.5.4 Shaft Type E1 – External Splines

The shaft type E1 is equipped with an external spline (DIN 5480 - W28 x 1.25 x 21 x 8f) for integration into a gearbox. The specifications of all customer-relevant interfaces of the E1-shaft are shown in Figure 27 as an example in combination with a S-flange.

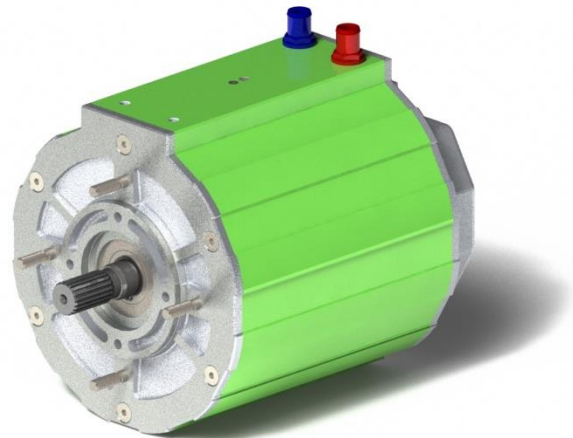


Figure 26: Shaft type E1 with flange type S1

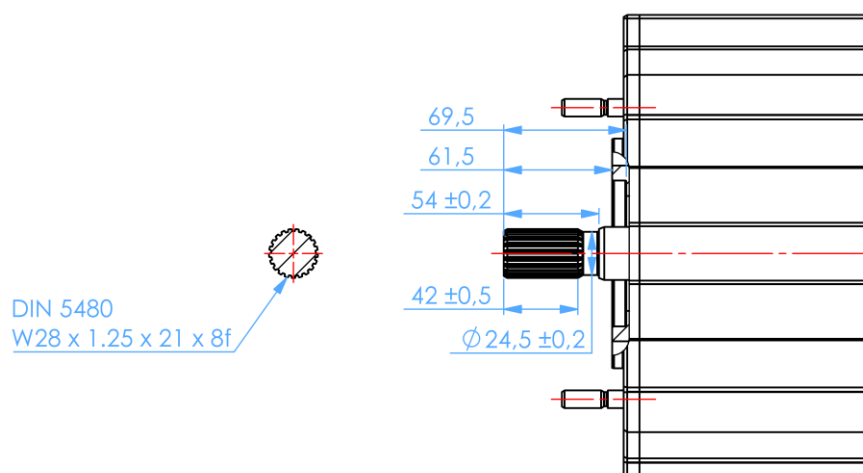


Figure 27: Dimensions shaft type E1

NOTICE



The motors are supplied with corrosion protection on the shaft end. This is not intended as a lubricant for the spline shaft. Spline couplings require specific lubrication according to the characteristics of the application and ENGIRO recommends the consultation of the lubrication suppliers to ensure proper selection and definition of re-lubrication intervals.

5.5.5 Shaft Type G1 – Internal Spline 13T

Shaft type G1 is typically combined with housing flange C1. These machines are suited to be coupled with a hydraulic pump with Flange SAE type B / ANSI 101-2 and shaft SAE Type B-B / ANSI 25-4. The internal spline corresponds to ANSI B92.1 13T 16/32DP 30° and is hardened. The specifications of all customer-relevant interfaces of the G1-shaft are shown in Figure 29 as an example in combination with a C1-flange.

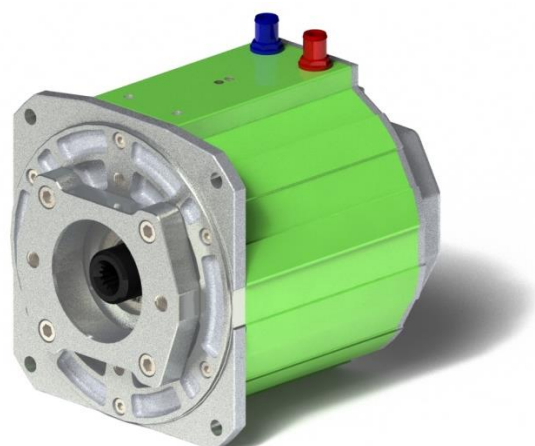


Figure 28: Shaft type G1 with flange type C1

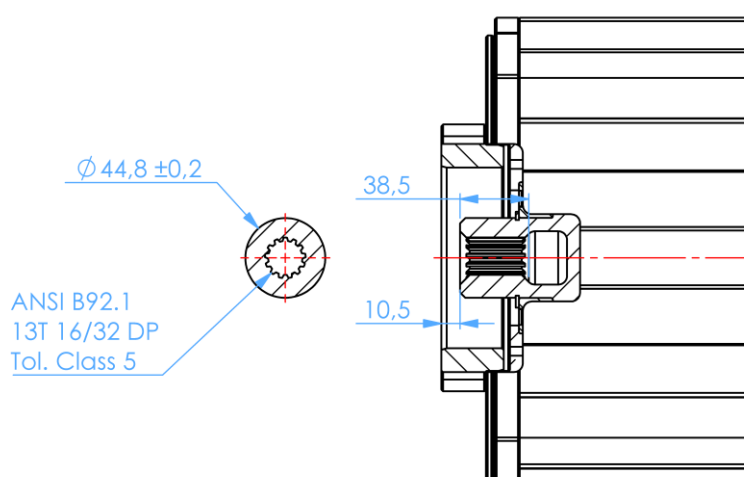


Figure 29: Dimensions shaft type G1

NOTICE



The motors are supplied with corrosion protection on the shaft end. This is not intended as a lubricant for the spline shaft. Spline couplings require specific lubrication according to the characteristics of the application and ENGIRO recommends the consultation of the lubrication suppliers to ensure proper selection and definition of re-lubrication intervals.

5.5.6 Shaft Type H1 – Internal Spline 9T

The shaft type H1 is typically combined with housing flange S1. These machines are suited to be coupled with a hydraulic pump similar to a Flange SAE type A / ANSI 82-2 (deviation on hole distance Ø108 vs. Ø106.4) and shaft SAE Type A / ANSI 16-4. The specifications of all customer-relevant interfaces of the H1-shaft are shown in Figure 31 as an example in combination with a S1-flange.

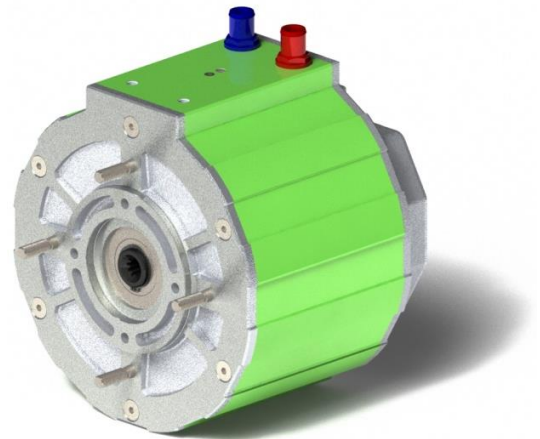


Figure 30: Shaft type H1 with flange type S1

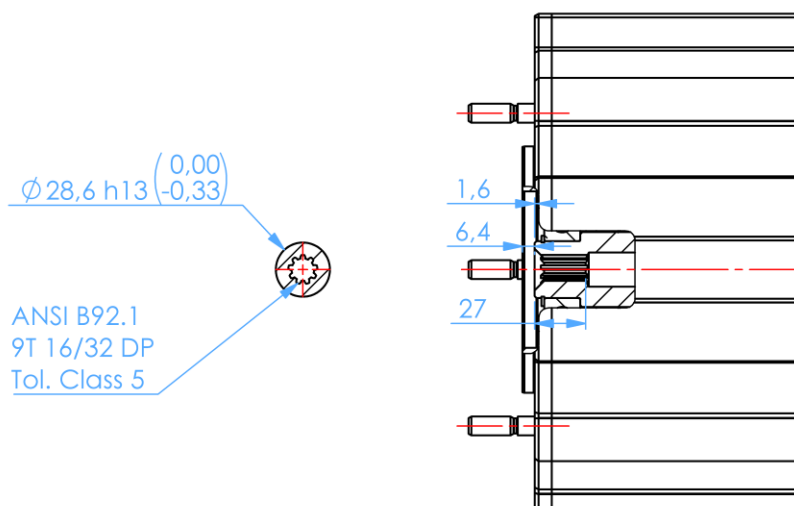


Figure 31: Dimensions shaft type H with flange type S1

NOTICE



The motors are supplied with corrosion protection on the shaft end. This is not intended as a lubricant for the spline shaft. Spline couplings require specific lubrication according to the characteristics of the application and ENGIRO recommends the consultation of the lubrication suppliers to ensure proper selection and definition of re-lubrication intervals.

5.5.7 Shaft Type J1 – External Spline 16T

The shaft type J1 is equipped with an external spline (DIN 5480 - W35 x 2 x 16 x 8h) for integration into a gearbox. The specifications of all customer-relevant interfaces of the J1-shaft are shown in Figure 33 as an example in combination with a D1-flange.

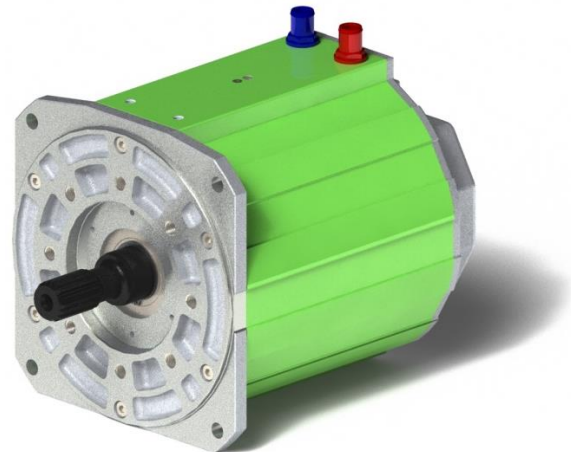


Figure 32: Shaft type J1 with flange type D1

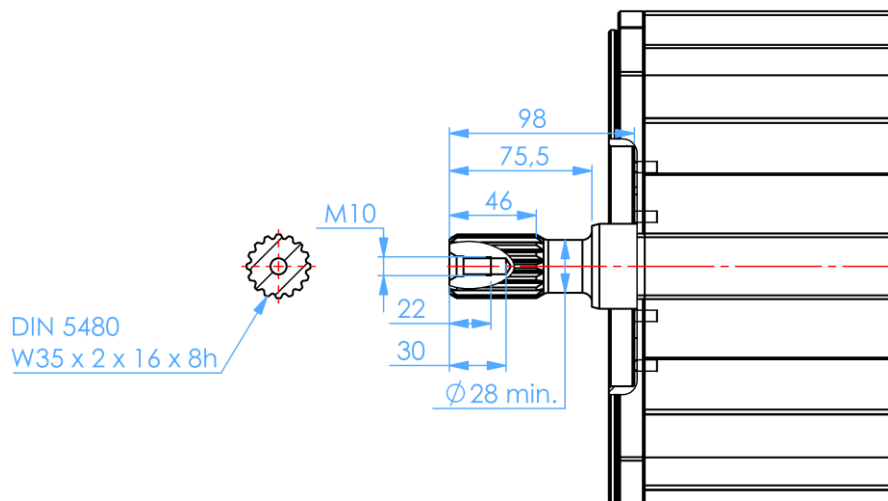


Figure 33: Dimensions shaft type J1

NOTICE



The motors are supplied with corrosion protection on the shaft end. This is not intended as a lubricant for the spline shaft. Spline couplings require specific lubrication according to the characteristics of the application and ENGIRO recommends the consultation of the lubrication suppliers to ensure proper selection and definition of re-lubrication intervals.

5.5.8 Shaft Type L1 – 35 mm Cylindrical Shaft

Machines with shaft type L1 are equipped with a cylindrical shaft of 35 mm diameter. The shaft has a 10 mm keyway and an M12 threaded hole. A hub can be either shrunk onto the shaft or be connected using a clamping element. Ensure that the connection is free of play and properly centred. The specifications of all customer-relevant interfaces of the L1-shaft are shown in Figure 35 as an example in combination with a D1-flange.

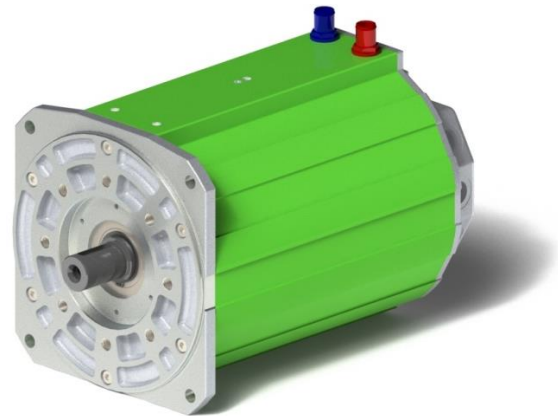


Figure 34: Shaft type L1 with flange type D1

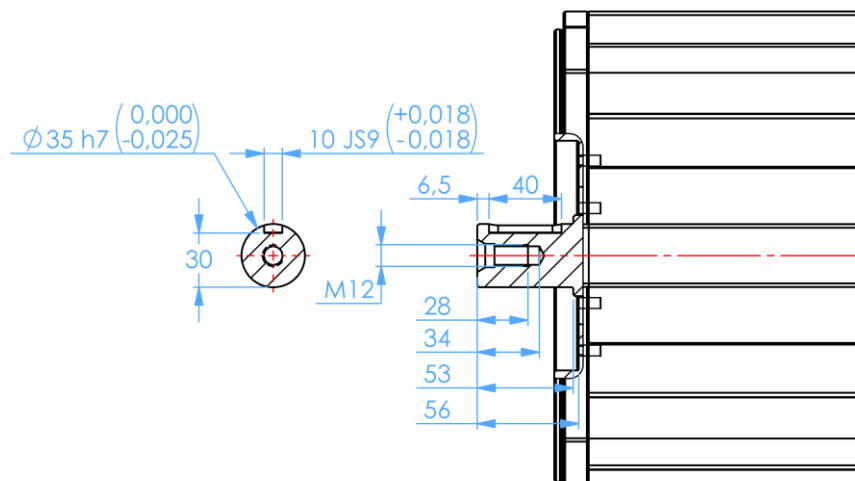


Figure 35: Dimensions shaft type L1



NOTICE

Please make sure that the mechanical load is distributed evenly along the length of the keyway. Avoid large torque impulses for shafts with keyways.

5.5.9 Shaft Type S1 – 28 mm Cylindrical Shaft

Machines with shaft type S1 are equipped with a cylindrical shaft of 28 mm diameter. The shaft has an 8 mm keyway and a M10 threaded hole. A hub can be either shrunk onto the shaft or be connected using a clamping element. Ensure that the connection is free of play and properly centred. The specifications of all customer-relevant interfaces of the S1-shaft are shown in Figure 37 as an example in combination with a S1-flange.

Key: DIN 6885 A 8 x 7 x 36

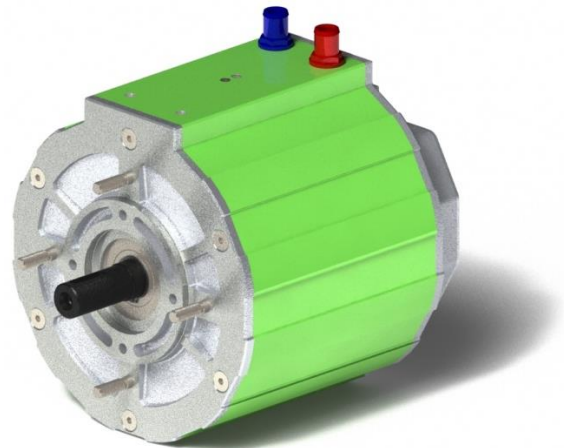


Figure 36: Shaft type S1 with flange type S1

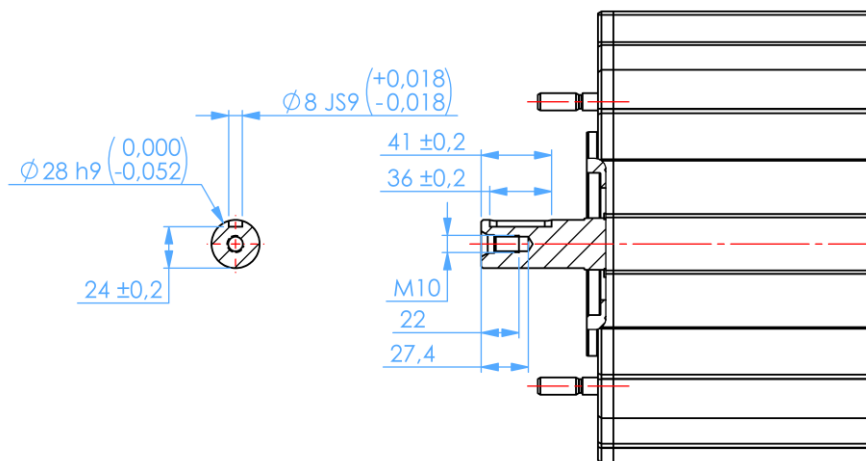


Figure 37: Dimensions shaft type S1



NOTICE

Please make sure that the mechanical load is distributed evenly along the length of the keyway. Avoid large torque impulses for shafts with keyways.

5.6 Possible shaft-flange combinations

Flange	Shaft
B1/C1/D1	C1/D1/G1/J1/L1
S1	E1/H1/S1

Please note that not all possible shaft-flange combinations may be available for a specific motor type. For more detailed information, please contact ENGIRO support.

5.7 Bearings

All standard 205W machines are equipped with two different non-isolated roller bearings (see 5.5.1). When the machines are operated with a power electronic inverter, bearing currents might occur in the machine. The phenomenon is among others strongly depending on the switching pattern of the inverter, the supply voltage as well as the load and the speed of the machine.

A suitable choice of isolated bearings and/or shaft grounding solutions is depending on the bearing current phenomenon and the system design. Taking uniquely measures at the machine might shift the issue to attached components, such as a hydraulic pump or a gearbox. ENGIRO can offer isolated bearings within the 205W machine series. Please contact ENGIRO to discuss the requirements of your application and the possible solutions.

5.8 External mechanical loads

ENGIRO advises to damp the motor from the chassis. This applies particularly for applications in which the motor is exposed to mechanical loads while not being operated, as vibration might damage the bearings due to lack of lubrication.

5.9 Impact Protection

Some applications (e.g., railway) may need an extra protection to prevent damages due to stone impacts to the motor. The customer is responsible to ensure such impact protection within the target application.

6 Cooling

6.1 IC Code

The 205W motor has a motor housing with an integrated cooling jacket for a cooling fluid (6.4). This primary cooling circuit is a water coolant mixture which circulates through the motor housing by an external water pump.

The secondary cooling circuit can be a cooling-fluid-to-air heat-exchanger (quite common in traction applications), resulting in the IC code IC9S7A7. Another option is a cooling-fluid-to-sea-water heat-exchanger (quite common in marine applications), resulting in the IC code IC9S7W7.

6.2 Cooling Fluid Pipes

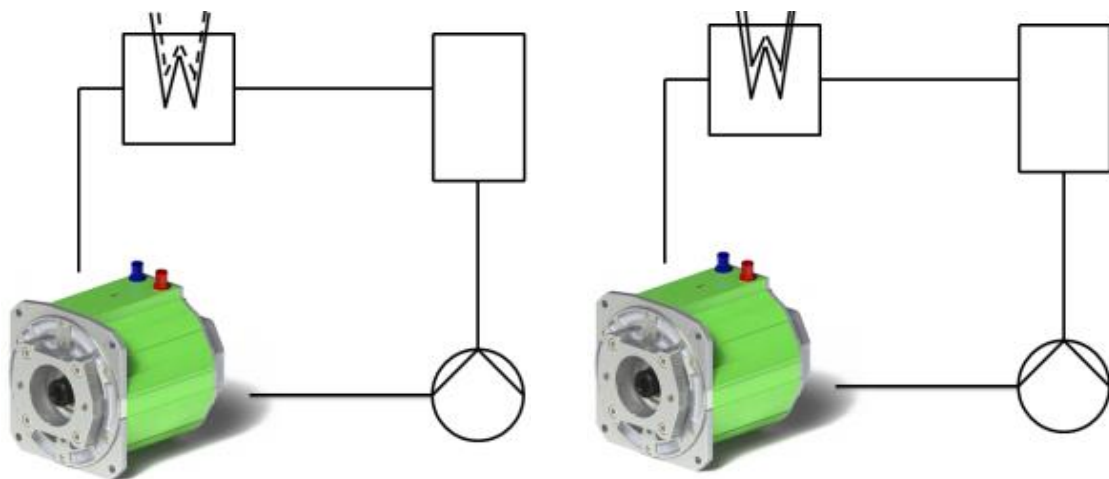


Figure 38: Liquid-to-air and liquid-to-liquid cooling

205W machines are designed for liquid cooling and, therefore, equipped with 19mm ($\frac{3}{4}$ ") hose barbs. The direction of the fluid flow is reversible. Make sure to install an appropriate pump to ensure a sufficient cooling fluid flow as specified in the machine datasheet.

Depending on the application, the power electronic controller can be air cooled or liquid cooled. When liquid cooling is used, the controller can have a separate cooling circuit, or it can be included in the cooling circuit of the machine.



NOTICE

Depending on the cooling channel diameter of the power electronic controller, the cooling fluid flow for the machine can be restricted. In this case, an appropriate bypass must be calculated by equation (4) and incorporated in the cooling system design.

$$d_{\text{bypass}} = \sqrt{(d_{\text{main}})^2 - (d_{\text{controller}})^2} \quad (4)$$

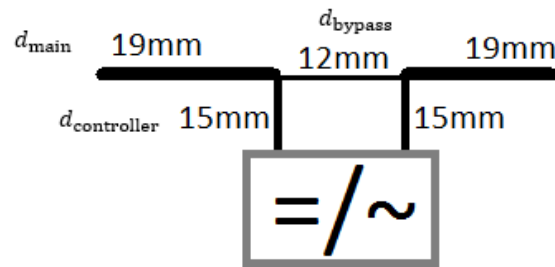


Figure 39: Example of a water bypass

6.3 Cooling and Mounting Position

For the cooling to function properly, the system must be bled. To ensure self-bleeding, the machine must be mounted such that no air can be trapped in the machines housing. Refer to section 5.1 for further information.

6.4 Cooling Medium

Aluminium corrosion and freezing preventing coolants should be used. ENGIRO advises to use OAT coolants, like G12/G12+/G30. Some examples:

- TL 774-D/F
- VIN 878389
- MAN 324 SNF
- MTL 5048

Mixing different coolants should be avoided. Be aware of dependence of cooling capacity on concentration of the solution. A typical mixture of water and additives is 50 / 50.

Do not use cooling mediums of the type IAT or HOAT! Cooling mediums of this type contain silicates, which can lead to a malfunction of the housing seals. **Avoid cooling mediums with the following identifications:**

IAT

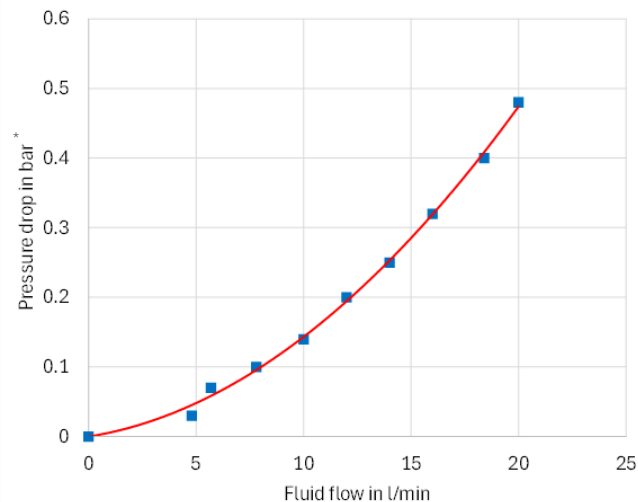
- G11/G48

HOAT

- G12++/G13/G40

6.5 Cooling Fluid Temperature, Flow Rate and Pressure

The machine performance is depending on the flow rate and the temperature of the cooling fluid. Make sure the installed pump can deliver the specified flow rate of the specific machine. The chosen radiator needs to be dimensioned to dissipate the machine losses at the specified flow rate and fluid temperature.



*Pressure sensor: total error 0.09 bar

Figure 40: 205W housing pressure drop over cooling fluid flow rate ($T_{\text{fluid}} \approx 23^{\circ}\text{C}$).



→ Consult the corresponding datasheet for the specific coolant requirements of a certain type 205W variant. The latest version of datasheets can be found here:
<https://www.engiro.de/en/products/motors-and-generators>

7 Electric Installation

Electric installation should be performed while following the basic rules:

- proper electrical contact where needed
- proper insulation where needed
- properly dimensioned wiring
- when applicable – shielded wiring with proper shield connections
- proper housing potential equalisation / equipotential bonding
- proper cable routing



NOTICE

Dirty, corroded or loose connections will cause extra heat and/or malfunctioning!

Refer to IEC 60204-1 for electric installation requirements.

7.1 Housing potential equalisation / equipotential bonding

In order to reduce electromagnetic disturbances and prevent uncontrolled currents, there must be proper equipotential bonding of the motor and the inverter.

Equipotential bonding is a low-resistance, low-inductance galvanic connection between the housings of the motor and the inverter. Basic rules for electrical connections as well as equipotential bonding connections apply (e.g. star connection, avoiding “ground loops” etc.).

The following example of an equipotential bonding is a proposal by ENGIRO and may be used by the Customer’s system integrator. The system integrator must ensure that the design, implementation and equipotential bonding meet the system’s environmental conditions. ENGIRO takes no responsibility for the design and implementation of the equipotential bonding as environmental conditions must be considered when designing and implementing the equipotential bonding.

ENGIRO recommends using a flat braided strap with approximately the same cross section area as a phase cable.

Equipotential bonding requirements of standards for a specific application (such as resistance and cross-section) might differ from the below mentioned standards and are the responsibility of the system integrator.

Cross-section of HV cable/phase connection	Minimum cross-section of equipotential bonding means, conductor or ground strap	Unit
$S < 4$	4	mm ²
$4 \leq S \leq 25$	S	
$25 \leq S \leq 50$	25	
$S > 50$	0,5 x S	



Figure 41: Equipotential bonding braided strap.

There are two options for the attachment of the bonding conductor to the motor housing:

1. M6 brass stud inside the rear flange
2. M10 threaded bore at the rear

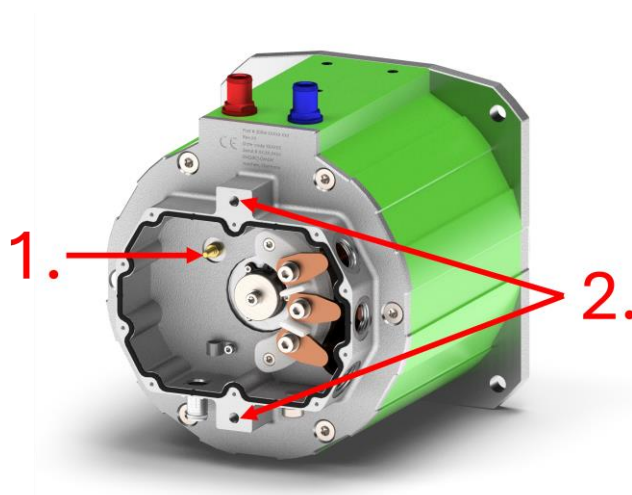


Figure 42: Options for a bonding strap connection.

Tightening torques:

1. M6 brass stud inside the rear flange: **4 Nm** ^{+/-4%}
2. M10 threaded bore at the rear: **31 Nm** ^{+/-4%}

When using the ground connection inside the motor connection box ENGIRO recommends using a nut with washer and split washer:

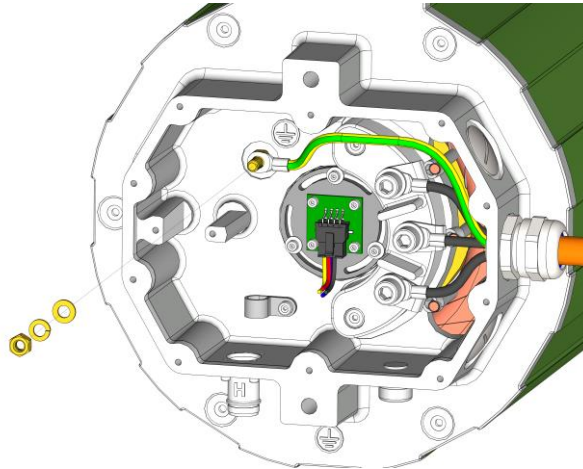


Figure 43: Ground connection inside the motor connection box

When using the M10 threaded bore at the rear flange for grounding connection ENGIRO recommends using an NSK-E10 safety washer with an M10 strength class 8.8. screw:

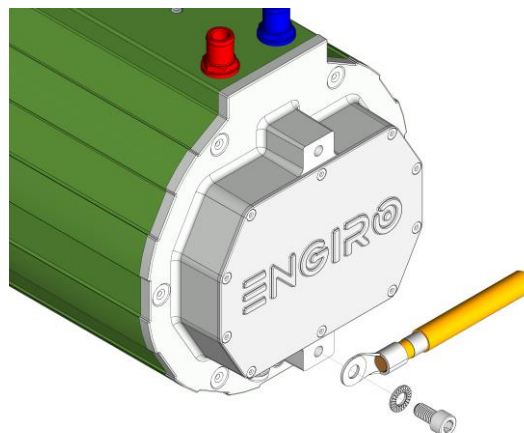


Figure 44: Ground connection using M10 threaded bore at the rear flange



NOTICE

The M10 threaded bore is not suitable to be used as ground connection and motor mounting **at the same time**.

For increased vibration requirements, additional screw locking to prevent loosening (e.g. with Loctite 243) of screws is recommended.

NOTICE

Mounting a motor and an inverter on the same (sub)frame often doesn't suffice as equipotential bonding. Therefore, ENGIRO recommends using a conductor for the connection.

High voltage inverters usually have dedicated threaded bores in the housing for such a connection, while low voltage inverters often do not. In such case, the cooling plate can be used for the connection as well.

7.2 Phase Cable Connection

The safety rules provided in chapter 2.2 must be followed!

To connect the phase cables, the rear cover with the ENGIRO logo must be removed.

For proper installation & disassembly of the rear cover for 205W-Motors with HVIL sensor, the following installation and disassembly instructions must be observed to prevent damage to the components. For motors without HVIL, the assembly/disassembly sequence in which the screws of the rear cover are loosened/mounted is arbitrary.



⚠ WARNING

With the cover removed, a large opening towards the active parts of the motor becomes uncovered. Pay special caution during unmounting of the cover, connecting the phase cables and closing the cover. Subjects or fluids dropped inside will result in damage, the motor will have to be returned to ENGIRO for disassembly. Do not leave the motor open unnecessarily. Never operate the machine without a properly installed rear cover!

Use 2.5 mm Allen key to loosen the 10 screws (Figure 45) according to the following instruction:

1. It is important to leave the screws marked ④ and ⑤ (directly above the HVIL switch) fitted until the end.
2. First remove all screws marked with ①.
3. Before removing the screws ②, ③, ④ and ⑤, the cover must be depressed by hand.
4. When all screws have been removed, the hand pressure can be slowly released, and the cover can be removed.

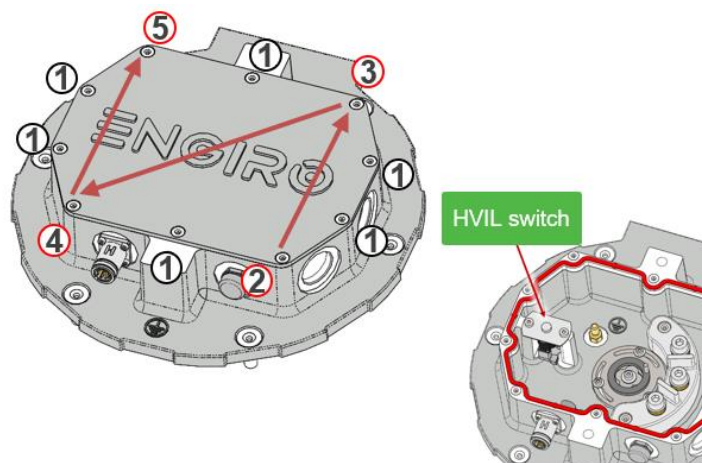


Figure 45: rear cover with screw numbering for rear cover disassembly

For the installation of the phase cables, screws with washers and thread lock coating (Variant A) (Figure 46) or screws with NSK-E washers (Variant B) (Figure 47) should be used. Mounting material included in the scope of delivery from Rev.14 (Variant A or B). The pre-assembled screws only secure the stator phase connections for storage and transport. Do not reuse these pre-assembled screws.

- **Tightening torque phase terminal** = **22 Nm^{+/- 1 Nm}**
- **Tightening torque cover screws** = **3 Nm^{+/- 0.1 Nm}**

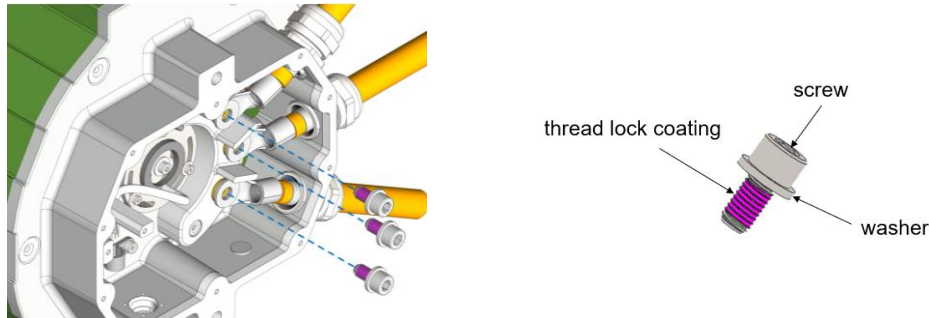


Figure 46: Phase cable terminals with screws with washers and thread lock coating (Variant A) screw connection



NOTICE

The thread lock coating is a high-strength, temperature-resistant microcapsule coating. For complete curing after tightening, the screw connection must rest for 24 hours. Do not reuse the screws! New coated screws are available from ENGIRO on request.

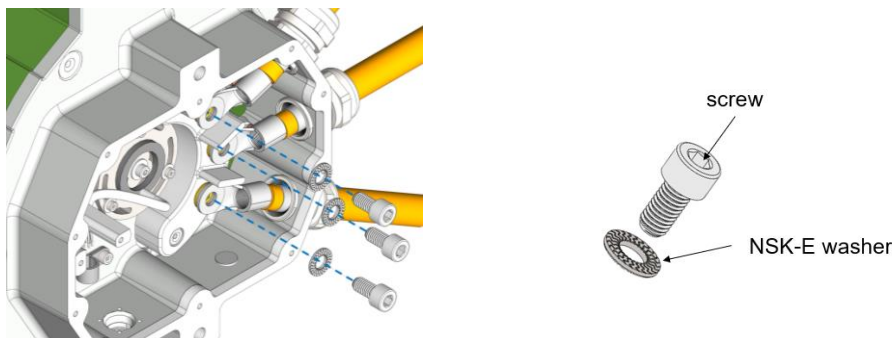


Figure 47: Phase cable terminals with screws and NSK-E washers (Variant B) screw connection

NOTICE



The NSK-E washers must be mounted with the detent edges facing the screw head as shown in Figure 47.

Do not reuse the NSK-E washers! New NSK-E washers are available from ENGIRO on request.

	Part Name	Engiro Part-No.
Variant A	(Set of 3) Screw acc. DIN 912 M8x16 St.8.8 galv. with Precoat 80 & washer	4880422
Variant B	Screw acc. DIN 912 M8x16 St.8.8 galv.	6237220
	NSK-E8 C60 Washer	6322518

Reinstall the rear cover after the phase cables are connected according to the following instruction.

1. Make sure the sealing is installed correctly.
2. To ensure that the HVIL sensor remains in the correct position, the terminal box cover must be pressed down (by hand) while tightening the screws ①, ②, ③, ④.
3. The following installation sequence, as shown in, must be observed ①→②→③→④.
4. All other screws (marked with ⑤) must then be fitted. The order is arbitrary.

For increased vibration requirements, additional screw locking to prevent loosening (e.g. with Loctite 243) of the terminal box screws (all) is recommended.

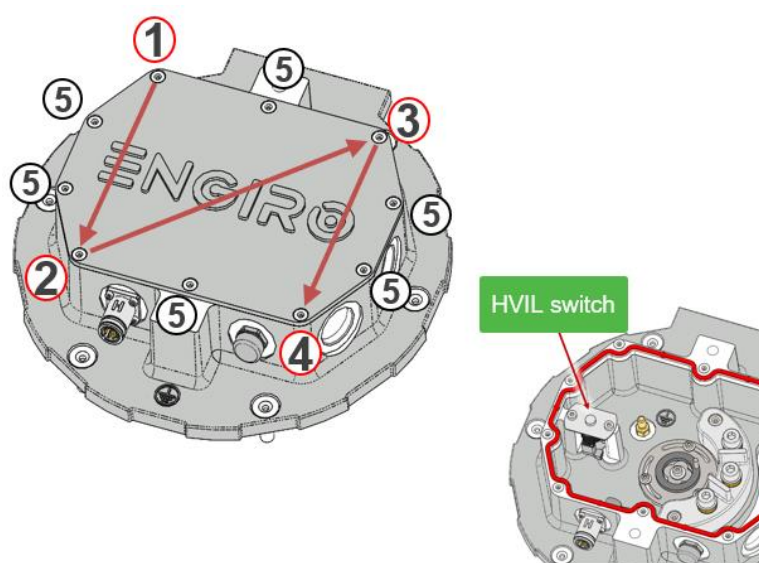


Figure 48: rear cover with screw numbering for rear cover disassembly



NOTICE

A right connection order of the phase cables is necessary for the machine to run properly. Wrong connection order results in unpredictable behaviour of the machine which may cause damage and/or injury. Consult section 8.1.

To ensure a proper electric connection of the 205W machines, for each cable diameter the proper cable lugs, crimping tool as well as cable glands must be used. The phase cable materials must be suitable for temperatures of up to 150 °C. Please refer to the following table for examples/recommendations.

	Copper cross section [mm²]	cable		cable gland		cable lug		crimping tool		crimp insert	
		type	article number	type	article number	type	article number	type	article number	type	article number
unscreened (< 140V DC)	25	RADOX 155 Orange 25 mm²		M20 unscreened 5-14 mm Reduction M20/25	Pflitsch bg 220ms Pflitsch 80.225/220	DIN 46234 25-8	Weitkowitz 41052	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90494
	35	RADOX 155 Orange 35 mm²		M20 unscreened 5-14 mm Reduction M20/25	Pflitsch bg 220ms Pflitsch 80.225/220	DIN 46234 35-8	Weitkowitz 41056	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90495
	50	RADOX 155 Orange 50 mm²	Huber+Suhner 85066291	M25 unscreened 11-20 mm	Pflitsch bg 225ms	DIN 46234 50-8	Weitkowitz 41060	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90496
	70	RADOX 155 Orange 70 mm²		M25 unscreened 11-20 mm	Pflitsch bg 225ms	DIN 46234 50-8	Weitkowitz 41065	AP 10	Weitkowitz 91120	Dornpresseinsatz DIN 46234	Weitkowitz 91471
(140 - 750V DC) screened	16	RADOX 155(S) 16 mm²	Huber+Suhner 84116032	M20 screened 9-14 mm Reduction M 25/20	Pflitsch bg 220ms tri Pflitsch 80.225/220	DIN 46234 16-8	Weitkowitz 41047	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90493
	25	RADOX 155(S) 25 mm²	Huber+Suhner 84100604	M20 screened 9-14 mm	Pflitsch bg 220ms tri	DIN 46234 25-8	Weitkowitz 41052	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90494
	35	RADOX 155(S) 35 mm²	Huber+Suhner 84100296	M25 screened 11-20 mm	Pflitsch bg 225ms tri	DIN 46234 35-8	Weitkowitz 41056	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90495
	50	RADOX 155(S) 50 mm²	Huber+Suhner 84096257	M25 screened 11-20 mm	Pflitsch bg 225ms tri	DIN 46234 50-8	Weitkowitz 41060	STILO1 45kN	Weitkowitz 91100	Dornpresseinsatz DIN 46234	Weitkowitz 90496
	70	RADOX 155(S) 70 mm²	Huber+Suhner 84100298	M25 screened 11-20 mm	Pflitsch bg 225ms tri	DIN 46234 70-8	Weitkowitz 41065	STILO20 60kN	Weitkowitz 91148	Dornpresseinsatz DIN 46234	Weitkowitz 91737

7.2.1 Recommended Cable Cross Section

It is recommended to choose the phase cables such that the current density is kept below 6 A/mm² at continuous operation and normal ambient conditions. Further guidelines on cable cross section can be found in many inverter manuals or in the application manual of the cable manufacturer. At voltages > 140V ENGIRO recommends shielded cables, or a rule of thumb: if inverter design allows usage of shielded cables, use shielded cables!

Exact sizing of cables is application dependent. A right choice of cable cross section and cable type is especially important when the (mounting) conditions are critical, e.g., extra thermal load because of a heat source like a combustion engine, vicinity of parts with low maximum temperature like some plastics, importance of weight in light-weight applications etc. Contact ENGIRO for advice when necessary.

7.2.2 Maximum cable length

To avoid excessive voltage losses and in some cases damage to the machine, it is necessary to use appropriately dimensioned power cables. In addition to the cable cross-section, the length takes an important role and should be determined in advance. If you need help finding the maximum cable length for your machine, please contact ENGIRO. Rule of thumb: as short as possible!

7.2.3 Cable Glands

The connection box has three **M25x1.5** threaded bores for cable glands. When an inverter is designed for use with shielded cables, it should be done. The shield must be connected to the machine and the inverter using shielded cable glands. Unshielded cable glands usually have larger inner bores which is convenient for low-voltage/high-current applications. Choose the right size cable glands with respect to the cable outer diameter to ensure high ingress protection grade and a tight fit for the cable.



Figure 49: Example of a cable gland

Some inverter manufacturers use multi-phase cables, where all three phase wires (and the **equipotential** bonding wire) are combined in one cable. If this cable does not fit in a standard M25 cable gland, suitable adapters are available. The remaining threaded bores must be closed with appropriate dummy plugs.

Appropriate adapters for single cable machine connection

M25 → M32	Phoenix Contact, 1647679, ENLAR-M-KV-M25/M32
M25 → M40	Phoenix Contact, 1647679, ENLAR-M-KV-M25/M32
	Phoenix Contact, 1647637, REDUC-M-KV-M32/M25
	Phoenix Contact, 1647679, ENLAR-M-KV-M25/M32
	Phoenix Contact, 1647682, ENLAR-M-KV-M32/M40
Dummy plugs	LAPP, 52103135, SKINDICHT® BL-M25 x 1, 5

If different adapters are used, make sure proper sealing is present or the ingress protection grade of the machine might degrade. Don't hesitate to contact ENGIRO for support.

NOTICE

To prevent loosening of the phase cable connections due to vibrations it is recommended to reduce the freedom of movement of the cables as far as possible.

Most cable glands are designed to provide a high-grade ingress-protected cable insertion into the motor, while not designed to provide a strong mechanical connection or strain relief. Proper cable routing is important to ensure the high ingress protection. Information about cable routing mechanics with respect to the glands and the cables can be found in their datasheets.

For cable installation requirements, please refer to EN 60204-1:2019 section 12.

7.3 Machine Sensors

All wires carrying sensor signals should be limited to a nominal current of 100 mA and a maximum current of 200 mA. Furthermore, these wires should not surpass a length of 5 metres.

7.3.1 Temperature Sensor

All 205W machines are equipped with a KTY 84-130 temperature sensor with the following resistance characteristics at 1 mA measuring current:

Temperature (°C)	Resistance (Ω)		
	Min.	Typ.	Max.
-20	403	424	446
-10	437	460	483
0	474	498	522
10	514	538	563
20	555	581	607
25	577	603	629
30	599	626	652
40	645	672	700
50	694	722	750
60	744	773	801
70	797	826	855
80	852	882	912
90	910	940	970
100	970	1000	1030
110	1029	1062	1096
120	1089	1127	1164
130	1152	1194	1235
140	1216	1262	1309
150	1282	1334	1385
160	1350	1407	1463
170	1420	1482	1544
180	1492	1560	1628

 WARNING



While the KTY sensor has a PTC-characteristic, it is not a PTC-resistor, and its polarity must be considered! Wrong polarity will result in wrong temperature measurement at critical temperatures, which will result in motor damage at high loads.

The resistance values depend on the measuring current. Some inverters deviate a lot from the standard measurement current. The resistance table must be adjusted accordingly to prevent wrong measurement!

7.3.2 Position Sensor Type E - Encoder

Machines with position sensor type E are equipped with an encoder position sensor.

Encoder position sensor (analogue sinusoidal)	
Power supply (VDD)	5 V $\pm$ 10 %
Power consumption	8 mA
Outputs	Signal amplitude pp: 2.5 V $\pm$ 0.5 V
Resolution	1 sine/cosine wave per revolution
Accuracy	Typ. $\pm 1^\circ$
Output Offset	50% VDD

7.3.3 Position Sensor Type R - Resolver

Machines with a position sensor type R are equipped with a resolver position sensor.

R Resolver position sensor	
Pole pairs	2
Excitation voltage	2.4 V _{rms} or 7 V _{pp}
Excitation frequency	10 kHz
Transformation ratio	0.5
Electrical error	$\pm 10'$ max
Phase shift	8°
Input impedance	100 $\pm$ 15 Ω
Output impedance	440 $\pm$ 66 Ω
Primary resistance	18 $\pm$ 1.8 Ω
Sine resistance	57 $\pm$ 5.7 Ω
Cosine resistance	57 $\pm$ 5.7 Ω

7.3.4 Position Sensor Type F - Resolver

Machines with a position sensor type F are equipped with a resolver position sensor.

F Resolver position sensor	
Pole pairs	4
Excitation voltage	7 V _{rms}
Excitation frequency	10 kHz
Transformation ratio	0.286
Mechanical error	≤ ± 30' max
Phase difference	≤ ± 15°
Input impedance	100 ± 15 Ω
Output impedance	161 ± 24 Ω
Primary resistance	22 ± 3 Ω
Sine resistance	31.5 ± 4 Ω
Cosine resistance	30 ± 4 Ω

7.3.5 Position Sensor Type N

Machines with a position sensor type N are not equipped with any position sensor.

7.4 Motor Sensor Cable

The machine is equipped with a sensor connector with male pins. Various sensor cable lengths can be offered by ENGIRO. For inverters offered by ENGIRO complete wiring looms with the suitable inverter connector are available.



Figure 50: Example of complete wiring loom.

In case of increased mechanical requirements (vibration loads) every thread of the Hummel sensor connector (the connector has several threads) should be secured with bolt adhesive. For further information please contact ENGIRO.

For self-assembly of a motor sensor cable the following parts are required on the machine side of the cable:

Connector housing for the cable:	Hummel, 7.831.500.000
	Hummel, 7.810.500.000
	Hummel, 7.811.500.000
Insert for female pins:	Hummel, 7.003.910.102
Female pins:	Hummel, 7.010.981.002
Crimp tool:	Hummel, 7.000.900.904
Recommended cable:	Lappkabel Unitronic LIYCY / TP / 4 x 2 x 0,5



NOTICE

ENGIRO provides no warranty for customer-made sensor cable. A faulty sensor cable might destroy the inverter and or the machine sensor. For correct cable assembly please refer to the following sections.

7.4.1 Shielding

The shielding of the sensor cable shall only be connected near the inverter, run up to the machine, but not be connected to the machine. Please always refer to the inverter manual. Depending on the inverter type, there is a dedicated pin in the logic connector, or it must be connected to 0V of the logic supply or 0V of the main battery for low voltage inverters. Example for the shielding can be seen in Figure 51 and Figure 52.

7.4.2 Encoder Cable

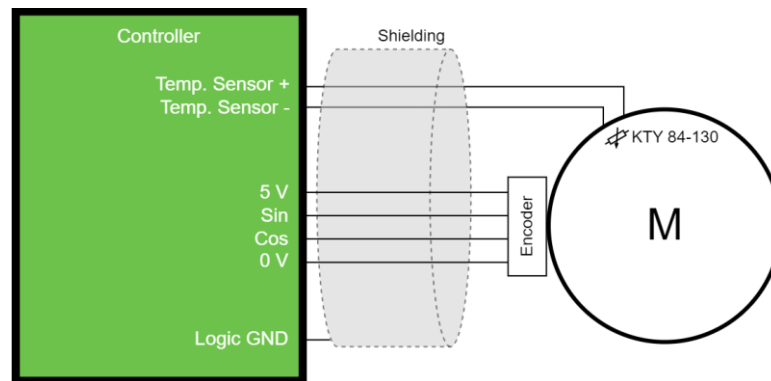


Figure 51: Encoder cable example

Sin- and Cos-signal wires should be a twisted pair. The encoder and the thermistor share the signal connector on the machine; thus, the sensor cable must accommodate two wires for the thermistor as well. For cables provided by ENGIRO, the Sin- and Cos-signals as well as the encoder supply and thermistor have twisted pairs.

Table 1: ENGIRO encoder cable signals

	signal	Hummel 10P connector
Twisted pair	KTY-	1
	KTY+	5
Twisted pair	Sensor_GND	7
	+5V	8
Twisted pair	Cos	9
	Sin	10

7.4.3 Resolver Cable

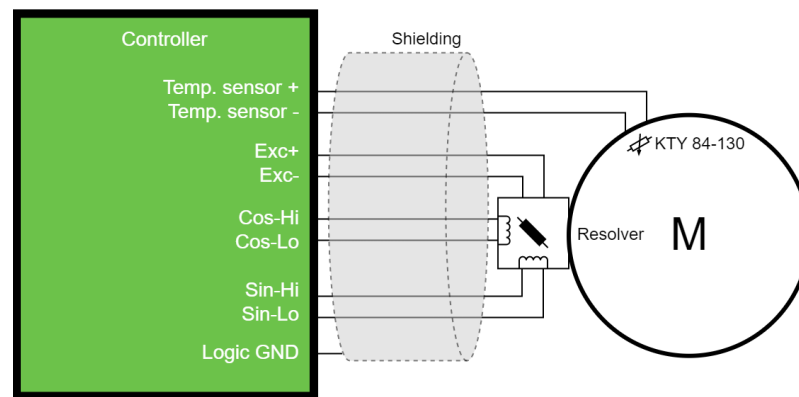


Figure 52: Resolver cable example

Sin - SinLo, Cos - CosLo and Exc+ - Exc- signal wires must be twisted pairs.

Some controllers accommodate multiple 0V pins on the logic connector, one of which can be used to connect shields. On controllers which do not have extra 0V pins, the shielding must be connected to a splice on the logic supply 0V.

For cables provided by ENGIRO, the Sin-SinLo, Cos-CosLo, Exc+, Exc-, HVIL and thermistor signals have twisted pairs.

Table 2: Resolver cable signals

	signal	Hummel 10P connector
Twisted pair	KTY-	1
	KTY+	5
Twisted pair	Sin-Hi	6
	Sin-Lo	2
Twisted pair	Cos-Hi	9
	Cos-Lo	7
Twisted pair	Exc+	3
	Exc-	4
Twisted pair	HVIL N	8
	HVIL P	10

7.5 Inverter Installation

7.5.1 Hardware

To minimize any EMC problems, general rules for high frequency installations apply. It is advised to mount the inverter as close as possible to the machine to minimize the lengths of phase- and signal cables.



Read, understand and apply all the safety precautions described by the inverter manufacturer. Insure proper protection against shock, moisture, temperature etc. Refer to the inverter manual for further information.

7.5.2 Software and Settings

ENGIRO offers pre-set inverters as well as (application specific) parameterization sets for customers with the possibilities to install the setting files themselves.




WARNING

No warranty for customer made or modified parameter files. An incorrectly parametrized drive set may cause serious damage and/or injuries! Always contact ENGIRO if an adjustment is needed.

The application/customer specific settings such as torque map, speed/torque control, throttle and switch settings etc. need to be discussed and defined upfront. Contact ENGIRO if you need support for a parameter setup.

7.6 Battery / DC-Supply Connection

Only qualified staff is allowed to work on systems with voltages above 60V. General rules and precautions for systems with high voltages / high currents apply. Always make sure all safety hints mentioned in section 2 are read, understood and followed by any staff working with high voltages. Always follow the inverter manufacturer's instruction manual. Never connect the DC power cables directly to the machine!

Most inverters on the market require a main contactor (e.g. low voltage inverters with internal pre-charge) or a pre-charge contactor as well as a main contactor (e.g. inverters without internal pre-charge). The contactor(s) must be dimensioned, mounted and supplied correctly to ensure safe operation.



WARNING

Malfunctioning contactor(s) may cause damage to the inverter and other electronics in the system.

Although DC cables are prone to cause less EMC issues than AC cables, it is advised to keep the cables as short as possible. Some manufacturers suggest putting a toroidal ferrite core around the DC cables in case of EMC issues. Make sure the ferrite is chosen for the right frequency range and able to handle the losses in continuous operation.

8 Commissioning and Operation

8.1 Initial Commissioning

Before putting the machine in operation please ensure the following steps are taken care of:

- Install machine and controller in the correct mechanical position (see chapter 5).
- Make sure the cooling is properly installed and working.
- Make sure all electric installation is carried out properly (see chapter 7)
 - o Please consider low-voltage and high-voltage cabling.
- Make sure the employed controller is parametrized accordingly.
- Depending on the control method (digital/analogue or CAN), make sure all required auxiliary devices are correctly installed and ready for operation.
- Verify that rotation- and torque direction are setup correctly for the application in order to prevent unexpected behaviour or damage to attached parts.
- Make sure all electric connections are protected against unintended contact.

8.1.1 Phase Cable Order and Rotation Direction

The phase cable order together with the controller parametrization determines the rotation direction of the machine / forward direction definition. ENGIRO provides three phase cable connection orders depending on the motor type: Standard 3-phase-arrangement with one terminal on each phase, 3-phase-arrangement with two parallel terminals and 6-phase arrangement with two independent windings. The standard ENGIRO connection order for 3-phase-arrangement and for a motor with two windings is shown in Figure 53. ENGIRO labelling follows requirements of IEC 60034-8:2007+A1:2014. Note that controllers may have different labelling. Table 3 lists commonly used labelling with standard 3-phase-arrangement, Table 4 lists commonly used labelling for motors with two windings and Table 5 lists commonly used labelling for motors with 3-phase-arrangement with two parallel terminals on each phase. Note that only the first option in each of those tables (printed in bold) meets the requirements of IEC 60034-8:2007+A1:2014 and is used and recommended by ENGIRO. Attach each phase according to the applicable table.

Table 3: Labelling for three-phase motors

Connector 1	Connector 2	Connector 3
U	V	W
M1	M2	M3

Table 4: Labelling for motors with two windings

Connector 1	Connector 2	Connector 3	Connector 4	Connector 5	Connector 6
1U	1V	1W	2U	2V	2W
U1	V1	W1	U2	V2	W2
M11	M12	M13	M21	M22	M23

Table 5: Labelling for three-phase motors with parallel terminals

Connector 1	Connector 2	Connector 3	Connector 4	Connector 5	Connector 6
U1-1	V1-1	W1-1	U1-2	V1-2	W1-2

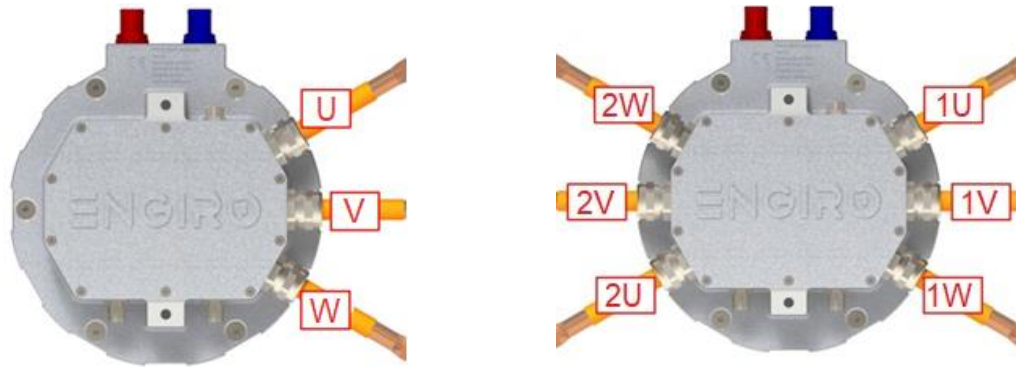


Figure 53: Standard connection order for 3-phase and two windings

The rotational direction of 205W motors is counter-clockwise (CCW) looking at the shaft end of the motor and following Table 3 according to IEC 60034-8:2007, as shown in Figure 54.

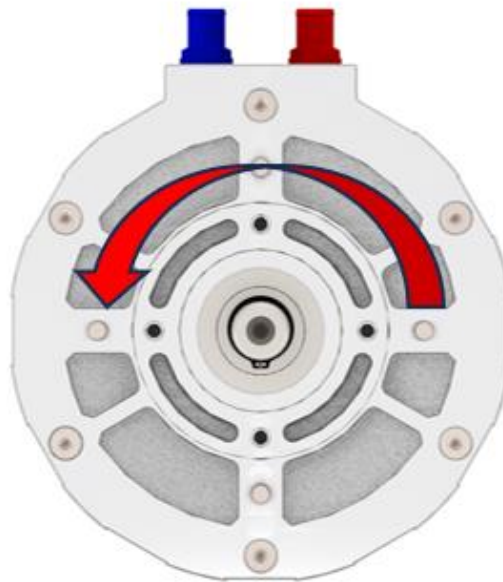


Figure 54: CCW rotation direction for ENGIRO motors

NOTICE



Various inverters have different rotation direction definitions and different ways of changing of the definition.

The phase order and therefore field rotation direction, sensor signal sequence and sensor offset must match. This is a combination of hardware connections and software settings.

A mismatch in connections or software would result in various malfunction: low torque and high losses, no rotation or uncontrolled spin-up ("runaway").

8.1.2 Vibration Limits

The machine is designed to meet the vibration limits stated in EN 60034-14. If you encounter unreasonably high vibrations, please make sure, the machine is properly mounted and all shafts in the system are aligned within the allowed tolerances. Another source of high vibration levels can be the parametrization of the inverter.

8.2 Operation

ENGIRO machines are intended to be integrated in larger systems, and ENGIRO is not responsible for the system design. The system designer must take care of a safe operation and the compliance with all applicable standards.



! WARNING

Ensure all safety features are connected and functioning properly. Do not bypass safety features! Contact ENGIRO when a feature needs to be modified.



! WARNING

Ensure the machine is properly attached. Torque changes might result in unexpected movements of improperly attached machines.



! WARNING

205W is a Permanent Magnet machine. This means that it generates voltage when spun. It has a mechanical rotational speed limit, and a variant-specific voltage-per-RPM ratio. Spinning the machine too fast might lead to various damage due to overvoltage, depending on the situation!

8.3 Temperature and Flow Rate Depending Performance

The power electronic controllers protect themselves and the machine from overheating. Overheating causes irreversible damage to machines and controllers, which can happen suddenly and cause damage to other components in the system or even the operator.

When reaching a certain temperature of the machine or the controller, the controller starts to reduce its output current to reduce thermal load. The temperature threshold is adjustable for the machine and usually predefined for the controller.

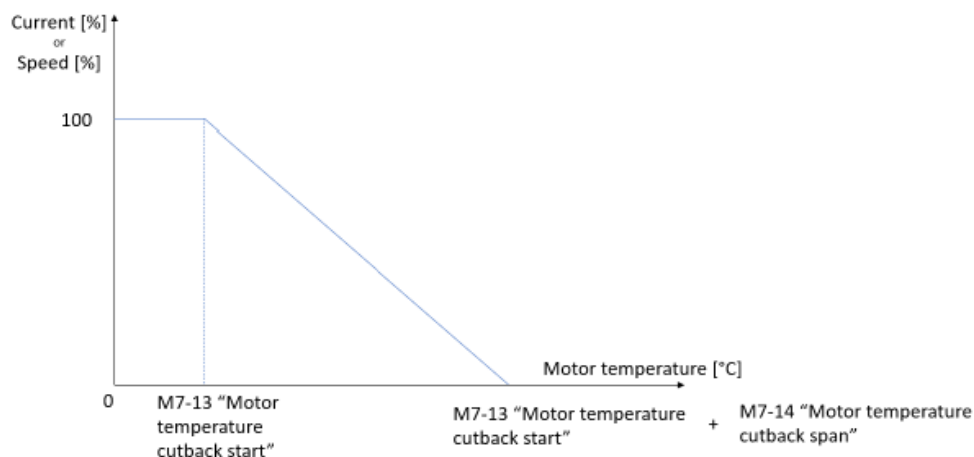


Figure 55: Example of temperature cutback from DMC SuperSigma2 manual

The temperature of the machine is measured at the coils/armature; the temperature of the controller is measured internally by the controller. To prevent derating, appropriate cooling must be guaranteed.



NOTICE

Always make sure that temperature protection is enabled and configured properly!

The “start of derating” for the machine temperature must be chosen such that the machine temperature will not rise above the defined Datasheet value.

Operating the machine/controller combination at its thermal limit negatively affects lifetime because of degrading material properties.

The motor and the inverter are designed for a certain coolant temperature and flow rate at which the right power output can be expected. Worse cooling will lead to faster derating and lower continuous power. Very bad cooling or no cooling at all can lead to damage of components!

8.4 Faults and Malfunctioning

Fault	Possible Reason	Solution
Temperature always high/low,	broken cable temperature sensor defect	Please verify if the measured resistance of the sensor is in line with the current machine temperature. Refer to chapter 7.3 for more information.
The machine oscillates when operated without load	If the machine is operated in speed mode, the speed control and gains might be improperly set for no-load operation	Adapt the speed control gains for the desired operation. Please contact ENGIRO if you need further support.
Machine rotates in wrong direction	Phase cable order incorrect.	
At very low torque levels, the machine accelerates uncontrollably and turns off with an over speed error	Phase cable and sensor cable order incorrect.	Please refer to section 7.4 and 8.1.1 for the correct installation orders. Contact ENGIRO if you need further support.
The machine does not rotate at all	Phase or sensor cable order incorrect.	
Controller switches off with sensor fault as soon as the output bridge is activated	incorrectly installed sensor cable shield and or ground cables	Verify all ground and shield connections at the machine and the inverter. Refer to section 7.1 for more information
	Improper cooling	Check the cooling system for a sufficient flow rate and temperature. Verify all connections and hoses for possible leakage.
Machine and or controller heat up much faster than expected	The machine and or controller are overloaded	Make sure the machine is operated within the specific limits stated in the datasheet
	Incorrect controller settings	Check the controller parameterization. Contact ENGIRO if the parameterization was included in the delivery

Fault	Possible Reason	Solution
The machine produces atypical noise	Improperly installed couplings.	Try to operate the machine without load. Contact ENGIRO if still noisy.
	Mounting screws are loose	Tighten screws. Replace damaged screws if needed.
	splines are worn out	Check splines visually. Contact ENGIRO if you have doubts.
	Improperly set inverter parameters	Check if the dataset installed on the inverter fits the specific machine. Contact ENGIRO if you have doubts.
	The machine is mechanically damaged	Please contact ENGIRO.

8.5 Inspection and Maintenance Interval

System Part	Inspection / Task	interval
Mechanical parts	Check for increased or atypical noise and vibration levels during operation	weekly
	Check for extensive dirt on any relevant machine part (e.g. shaft, housing, bearings, venting plug etc.). Clean the respective parts if necessary.	monthly
	Verify the tightening torque of all relevant mounting screws and covers ¹⁾ . Re-tighten screws if necessary. Replace broken screws. Contact ENGIRO if threads are damaged	yearly
	Check if the pressure venting plug is correctly installed. Clean if necessary.	monthly
Electric parts	Check cable for wear. Replace damaged cables. ¹⁾	monthly
	Check electric connections for correct tightness. Contact ENGIRO if any threads are damaged. ¹⁾	yearly
	Check grounding connections. Replace connections if necessary. ¹⁾	yearly
Cooling	Check for sufficient flow rate and cooling fluid temperature during operation.	weekly
	Check for fluid leakages. Re-tighten connections and replace if necessary.	monthly
	Check the quality of the cooling fluid medium. Replace medium if degraded. Flush the complete cooling system at high flow rate to remove sediments or particles.	yearly

1) In case of strong external stresses (e.g. corrosive environment, excessive vibration) inspections need to be carried out more frequently. Contact ENGIRO for further information

8.5.1 Cleaning Remarks

Only use non-abrasive, non-corrosive cleaning products compatible with aluminium. When a pressure hose is used to clean the machine, make sure the water jet is never directly pointed at a sealing (e.g. cable glands), bearings or the pressure venting plug.

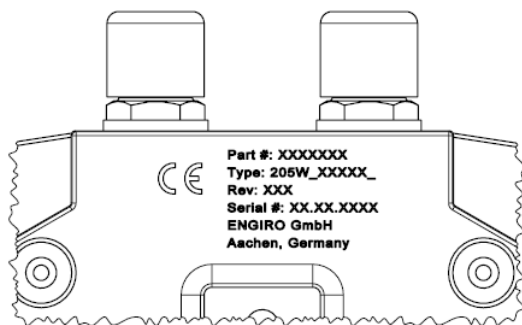
9 Service and Support

For spare parts and support, please contact ENGIRO or your local sales representative. Latest contact information can be found here: <https://www.engiro.com/kontakt>.

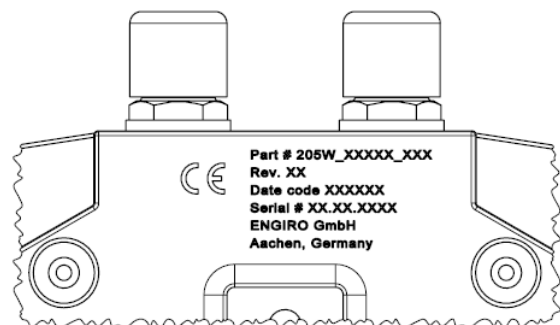
To provide you the best possible support, the following information is required:

- Exact part number of your machine
 - **After SAP transfer:** 7-digit number (see Figure 56, left)
 - **Before SAP transfer:** Starting with “205W_” (see Figure 56, right)
- Motor type
- Revision number of your machine starting with “Rev.”
- Serial number of your machine
- Point of purchase
- Information on the required spare parts and or support topic

Part and serial number are printed on the rear flange or on top of the motor housing depending on the revision of the machine as shown in Figure 56.



*Motor label **after** SAP transfer*



*Motor label **before** SAP transfer*

Figure 56: ENGIRO machine information

10 Disassembly and Disposal

In this chapter you find information about the disassembling and disposal of ENGIRO machines. Please follow local regulations for disposal.

10.1 Demounting and Disassembling

The demounting of the machine must be performed or supervised by qualified personnel (please refer to section 2.4). As the machines are intended for many purposes, ENGIRO only provides general demounting guidelines. Please take into consideration all demounting guidelines provided by the operator or manufacturer of the complete system.

Generally, the following steps must be performed:

- 1) Follow all safety rules in section 2, especially the five safety rules for work on electric components.
- 2) Disconnect all electric connections and remove all cables.
- 3) Remove all fluids such as lubricants and cooling fluids. Collect the fluids separately to dispose them according to local regulations.
- 4) Loosen the mechanical connections and unmount the machine.
- 5) Transport the machine to an appropriate place for further disassembly.



WARNING

The machine consists of very heavy parts which might fall.

- ➔ Always transport the machine with appropriate tools. Secure any parts of the machine against falling.
- ➔ Never transport the machine with less than 2 persons
- ➔ Wear appropriate safety equipment.

10.2 Disposal



Information

90% of the materials used in our machine are recyclable. For disposal and recycling please follow local regulations.

Alternatively, the machine can be returned free of charge to ENGIRO GmbH.



NOTICE

The disposal of the oil and detergents is accurately ruled by environment protection laws and environmental regulations.

- Deliver waste and used up lubricants to your hazardous waste point of acceptance.
- Slobbered lubricants must be fixed using binding agents and disposed of as hazardous waste.
- Make provisions to collect slobbered lubricants (sealed floor, collecting drip pans, collecting canvas covers).

11 Technical Data

11.1 General Data



Consult the corresponding datasheet for the specific data of a certain type 205W variant. The latest version of datasheets can be found here: <https://www.engiro.com/products/motors-and-generators>

11.2 Dimensions

A precise 3D-CAD model of each machine variant is available upon request. Latest contact information can be found here: <https://www.engiro.com/kontakt>.

To provide you the correct machine model please communicate the exact 7-digit part number of your machine.

12 Equipment and Replacement parts

To integrate the 205W into your application, ENGIRO offers various flange and shaft adapters. A list of readily available adapters can be found in section 12.1. If you need a different adapter, please contact ENGIRO for further assistance.

If parts of the delivery got lost or broken, please refer to section 12.1.2 for replacement parts.

12.1 Equipment

Part	Part number
SAE-4 Flange	4843024
data cable	See section 12.1.2
	-

12.1.1 Flange Adapter

To connect your machine via a standardized coupling to another device with a corresponding flange it is possible to mount a SAE-4 flange as shown below. For further information on how to install the flange return to section 5.4.

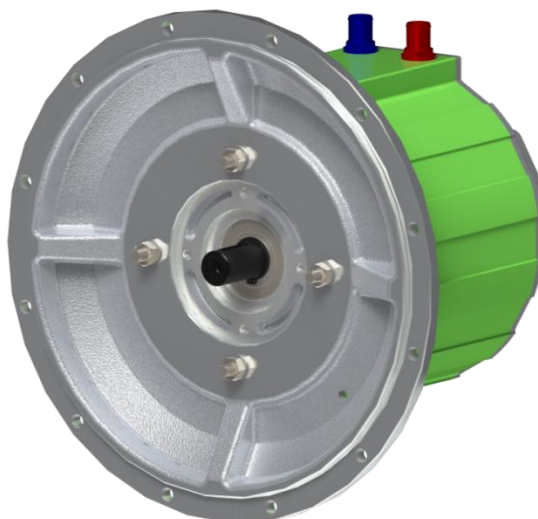


Figure 57: SAE-4 Flange (Part.-No.: 4843024).

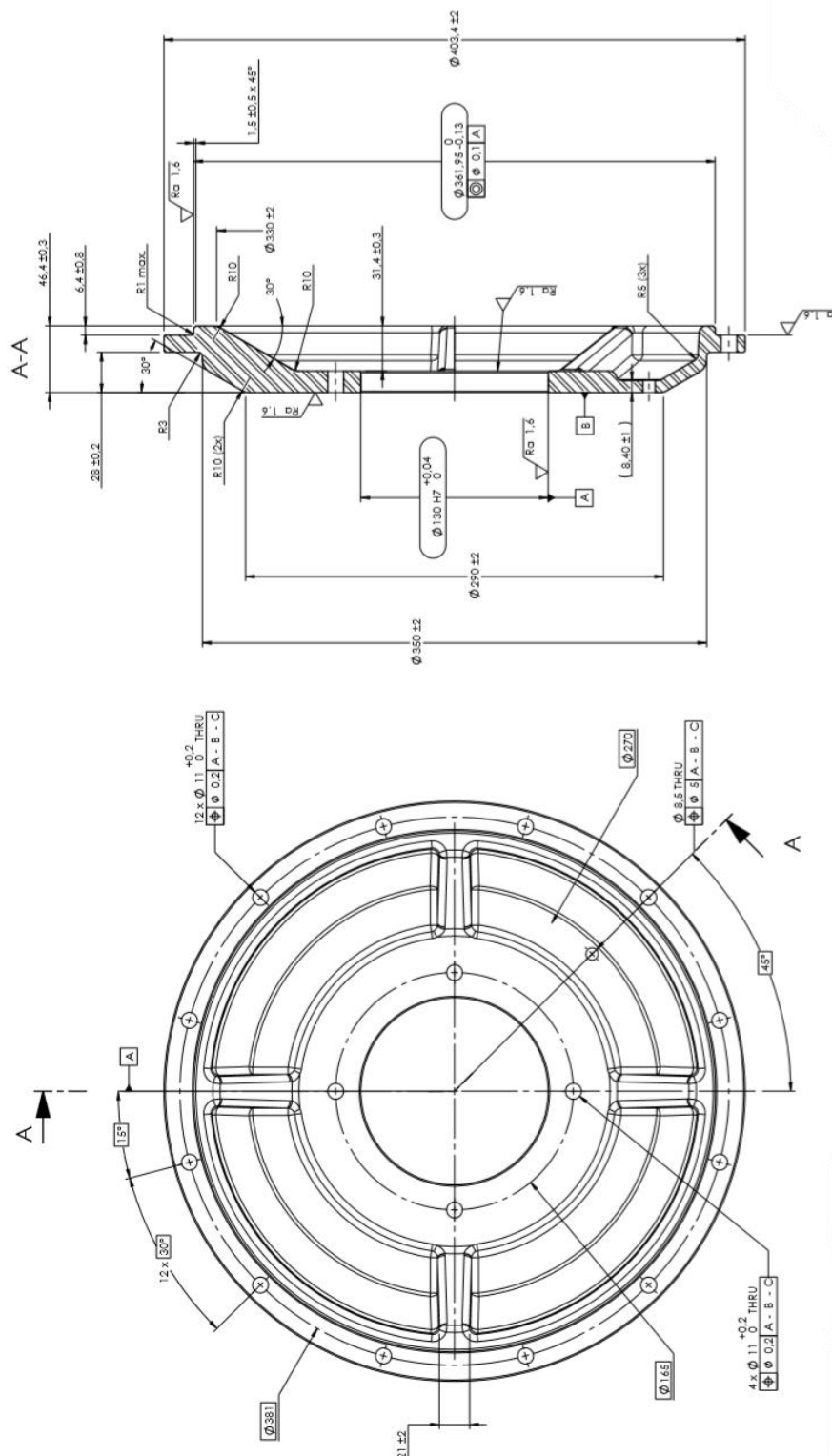


Figure 58: SAE-4 Flange (Part.-No.: 4843024).

12.1.2 Data cable

ENGIRO offers various data cables to suit different applications. These cables always include an assembled Hummel data connector on the machine end and open wires on the other end.

Part Description	Length	Part number
Single shielded resolver cable with 10 wires to connect Resolver and 2 temperature sensors to an inverter	2 m	4842833

If your application or inverter requires for a specific cable type, please make sure that the cable can be accommodated by the data connector used on the machine side (please refer to section 7.4). If you need assistance, please contact ENGIRO.

12.2 Replacement parts

Part	Part number
Mounting studs M10 x 35 for S1 flange	4842978
Sealing water connector 20x2	4842985
Water connector red M18x1,5	4842987
Water connector blue M18x1,5	4842986
90° Water connector M18x1.5 (machine-side)	4700852
90° Water connector (hose-side)	4700853
Sealing connection box	4842974

12.2.1 Flange S1 Machine Mounting Studs

The threaded mounting studs of flange type S1 can be replaced by standard parts according to DIN 939 with the use of thread locking (Loctite glue). Make sure you are using the exact parts specified in the table below to avoid any machine damage. The shorter thread of the studs has to be inserted in the machine.



Figure 59: Flange S1 machine mounting studs (Part.-No.: 4842978).

Machine Mounting Studs for Flange Type S1	
Type	DIN 939 – M10 x 35 – 8.8
Thread locking	Loctite glue (e.g. Loctite 243)
Tightening torque	20 Nm



NOTICE

When the studs are removed, care must be taken not to damage the machine windings. Any tool extending deeper than 20 mm in the threaded hole may damage the windings. Make sure no chips or other dirt falls into the machine while there are no studs installed.

12.2.2 Water connector

The Water connectors support cooling hoses with an inner diameter of 19mm. The connectors thread type is **M18x1.5**. To exchange your connectors please contact Engiro.



Figure 60: Water connectors (Part.-No.: 4842987/4842986).

For some applications it might be necessary to provide a cooling connection with a 90° outlet. For these purposes ENGIRO offers appropriate connectors, which additionally have a quick-engage function.



Figure 61: 90° Water connector (Part.-No.: 4700852/4700853).

12.2.3 Sealing

In case of leaking, the seals of the machine can be changed. Make sure the new seal has the following measurements and are resistant against the used cooling fluid (see section 6.4).

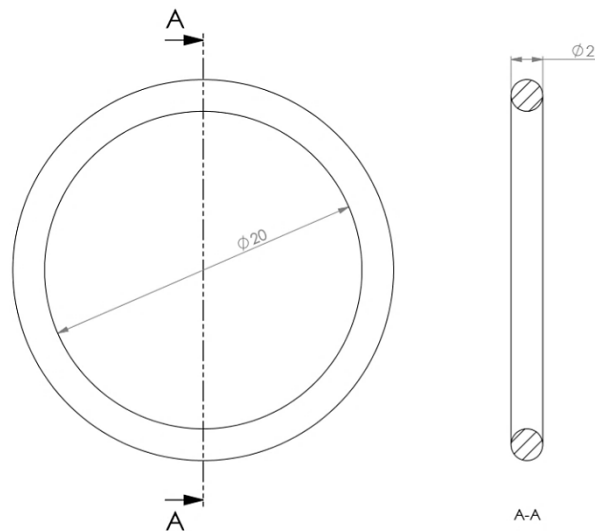


Figure 62: Water connector seal (Part.-No.: 4842985).

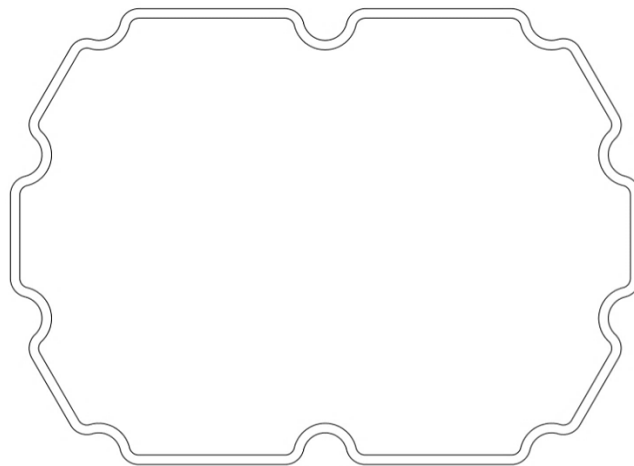


Figure 63: Rear cover seal (Flat gasket) (Part.-No.: 4842974).



CAUTION

If seals are replaced, make sure the connections are properly tightened, or cooling fluid might leak and damage equipment.

13 Appendix

13.1 CE-Certificate of Conformity

A CE certificate of conformity is available upon request.

13.2 ISO 9001

ENGIRO Quality

Within the scope of the entrepreneurial duty of care with regard to the quality assurance of our products and services, we hereby establish our quality policy.

Strategy

Our quality management system is a description of the requirements we have defined. Its application ensures that all activities that have an impact on business performance are planned, managed and monitored and that contractually agreed requirements are met.



Innovation requires reliability

The high quality of our drives and generators as well as the comprehensive services, we offer our customers, is the basis of our successful business activities. To us, quality means meeting the demands of our customers and interested parties in every respect. In this sense, quality is an important criterion for our daily business and forms a reliable relationship with our customers.

Ready for the future

We are committed to continuous improvement and further development of the QM system. The continuous improvement of the company is therefore a prerequisite for an effective reduction of costs and an important contribution to protecting the environment in the medium and long term. A smoothly running organization and the most up-to-date methods of quality management provide the necessary framework for this.

Internal awareness

All employees are obliged to apply the regulations valid in their area of responsibility, to constantly monitor their effectiveness and to adapt them to the latest knowledge and requirements. This understanding and awareness of quality as well as the mindset of all employees regarding quality are prerequisites for the satisfaction of our customers and thus for the lasting success of our company.

Our promise

By this declaration, the management obligates all employees and suppliers to carry out their activities in accordance with the descriptions of our QM system in order to ensure that the quality of all our company's products and services complies with internal and external requirements.

Sustainability

To us, acting quality-conscious also means using valuable resources efficiently and economically. All employees have the duty and right to seek to ensure that circumstances limiting the results of the company's performance are eliminated. <https://www.engiro.com/iso-9001-1>